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Software Development Group Project

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Software Requirement Specification Report

Project Title:

"Cataract Analysis using Smart Phone Images"

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Declaration Page

We hereby certify that this project report and all the artifacts associated with it are our own work and neither it has been submitted before nor is currently being submitted for any degree program.

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Abstract + Keywords

Abstract

A cataract is caused by the evaporation of the eye's normally clear lens. Cataract sufferers may notice that they can see through their clouded lenses, which can be disorienting. investigating through a fogged-up, icy window Obscure vision It will be more difficult to navigate websites because of cataracts. drive a car (particularly at night) or see the expression on someone's face a picture of a friend's face The majority of cataracts develop slowly and are unobtrusive. Observe early. Cataracts, on the other hand, can become a problem over time. with the ability to see There are some manual methods for detecting a virus. cataract. Blurred vision, faded colors, difficulty seeing in strong light, and other symptoms are the most common signs of this disease. Most of the time, these symptoms cause difficulty in performing various tasks. This disease affects 33 percent of the world's population. Disease detection is now a time-consuming process. Most people who are afflicted with this condition must see a doctor to determine whether they have it. Currently being tested Cataract detection from the retina is expensive when employing a fundus camera, and the procedure is not portable and requires an expert to conduct. This study describes a low-cost smartphone with smart system integration and micro lenses that allows patients in remote places to have frequent eye exams and the creation of novel disease diagnosis methods. The official mobile diagnostic system for analyzing retinal images captured with a tiny lens to diagnose retinal illness.

Keywords

Algorithm	A finite sequence of well-defined instructions.
Backend	The part of the system that is not directly accessed by the user.
Cataracts	A medical condition which resulting in blurred vision.
Dataset	A collection of data that is composed of separate elements.
Fundus camera	A specialized microscope with an attached camera designed to photograph the interior surface of the eye.
Image processing	The analysis and manipulation of a digitized image.
Ophthalmologist	A doctor who specializes in eye and vision care.
Ophthalmoscope	An instrument for inspecting the retina and other parts of the eye.
Supervised learning	A machine learning method in which machines are trained using labelled data.
Unsupervised learning	A machine learning method in which machines are trained by analyzing unlabeled data.

Table 1 : Table of Keywords

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List of Abbreviations

BCS	British computer society.
CNN	Convolutional neural network.
CPU	Central processing unit.
DSLR	Digital single-lens reflex.
GPU	Graphics processing unit.
GUI	Graphical User Interface.
ML	Machine Learning.
OOAD	Object-oriented analysis and design.
RAM	Random access memory.
SDLC	Software Development Life Cycle.
SRS	Software requirements specification.
SSADM	Structured Systems Analysis and Design Methodology.
UI	User interface.
UV	Ultraviolet.

Table 2 : Table for Abbreviations

Chapter 01 - Introduction

1.1. Chapter Overview

Cataracts are progressive cloudiness of the crystalline lens of the eye. The cloudy crystalline lens blocks and reflects the light that enters the eye, causing a variety of visual symptoms. Cataracts are the leading cause of vision loss in people over the age of 40 and the leading cause of blindness worldwide. They mature at different rates in different people but are often associated with aging. This chapter provides a background to the project to give a clear understanding of the problem domain, the challenges of existing solutions, the purpose and scope of the project, and the motivation to run the project in this area. This chapter also describes the resource requirements for the proposed prototype.

1.2. Problem Background

1.2.1. Problem Definition

Cataract is a significant global problem, particularly with the aging population and a lack of knowledge about it. With the growing demand for highly effective and cost-effective cataract extraction, health authorities are confronted with a major challenge. Cataracts are a clouding of the lens of the eye that normally affects vision. Cataracts, the most common cause of blindness and low vision, are often associated with aging. Sometimes children are born with the disease, or cataracts can develop from an eye injury, or from inflammation or other diseases, such as glaucoma and diabetes. Sometimes the development of cataracts is related to steroid use or can develop after exposure to certain types of radiation. Cataracts are responsible for more than 51% of blindness in the Eastern Mediterranean. Although cataracts can be easily treated surgically, access to eye care is limited in many countries. As people around the world live longer, the number of people with cataracts is expected to increase. Cataracts are also a major cause of visual impairment. (WHO EMRO | Cataract | Health topics, 2021)

1.2.2. Related Work

Up to date there have been several attempts and researches done in trying to create applications, and software aided medical tools in the improvement of cataract detection some of the works done are, (Pathak and Kumar, 2016) proposed an automated cataract detection algorithm using diagnostic opinion-based factor, the purpose of the study was to create a cheap and easy to use algorithm which could be employed by suitable devices which would be able to analyze and identify the existence of cataract from the true color photographs of an eye. The system functioned by recording characteristics such as the uniformity, intensity and standard deviation which would be computed and mapped with the assistance of a trained eye specialist to achieve the limitations and results of the proposed system. A system proposed by (Prasanna et al, 2013) where a mobile phone would be attached directly to ophthalmoscope, and the mobile phone would be used to take fundus images as seen directly through the ophthalmoscope the performance of the algorithm was test run using an existing database, the goal of the research was to create an easy hand-held, low-cost design which could be used by general practitioners and in health care sectors.

1.2.3. Problem Justification

Figure 1, shows that age is another factor needed to be taken into consideration, from 2005 to 2007, a study was undertaken on persons over the age of 60 in two areas in India's north and south regions, in which a total of 7518 people, 3586 from the north and 3932 from the south, were studied and diagnosed for cataract. It was determined from the findings that elderly adults in north and south India had greater incidence of unoperated cataract and Women had higher rates of cataract than men, which could not be explained by differences in surgical access. (Vashist, 2011).

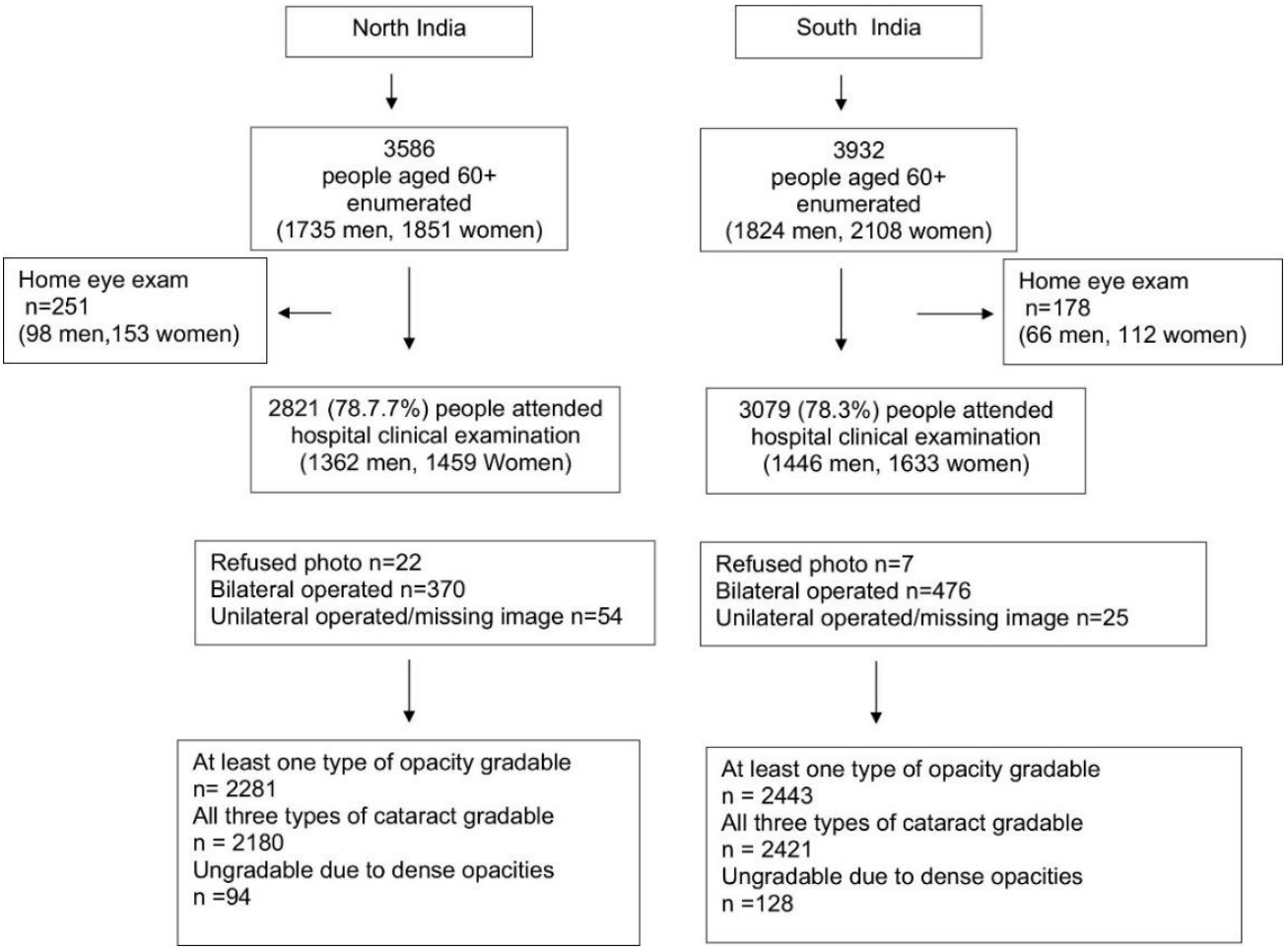


Figure 1 : Study to determine cataract at age 60 in North and South India by (Vashist, 2011).

1.2.4. Feature Comparison Table

Features	Our products	Product 1	Product 2	Product 3	Product 4	Product 5
Capture the eye of the patient using smartphone.	✓	✓	✓	✓	✓	✗
Test for cataract	✓	✗	✓	✓	✓	✓
Analyze the stage of cataract and review a list of symptoms	✓	✗	✓	✗	✓	✓
Evaluate the Mobile-based intelligent solution	✓	✗	✗	✗	✗	✗
Diagnosis Report	✓	✗	✓	✗	✗	✗

Table 3 : Feature Comparison Table

1.3. Problem Statement

Availability of Technology for the Diagnosis of cataract is not an easily accessible option in Rural areas of Sri Lanka.

1.4. Research Gap

1.4.1. Lack of research about Cataract patterns in Sri Lankans.

The main reason for Cataracts in human eyes is a direct result of ultraviolet (UV) radiation overexposure. Most of the research papers that have been published are more focused on the North Atlantic and Pacific regions. The temperatures and climates in these two regions are vastly different from the environments in Sri Lanka. Hence, the data these researchers use cannot be compared with Sri Lanka.

1.4.2. The accuracy of the datasets could be improved.

A lot of research papers on the internet were written in around 2019-2020. But the technology has moved on quite a bit from those times. New algorithms have emerged in the industry, and they are far better at computer vision than the algorithms of the past. And with more specialized datasets available in the market, there is a possibility to improve the accuracy of the models by using more accurate data.

1.4.3. Hardware requirements could be lower.

In the past research papers almost all of them used Iris scans as the dataset for the prediction process. This makes these programs inaccessible for the public. But in this project, there is a possibility to use simple images from a smartphone camera to do the prediction. This will require a more complex algorithm. And the approach won't be as accurate as the Iris scans, but the easy accessibility makes the difference.

1.4.4. The methods for training the models could be better

The most common methodology of train machine learning models is unsupervised learning. But with that method the accuracy or the program gets significantly lower. So, our project plans to train models using the supervised learning method. This way will be more time consuming to train the models, but the program will be more accurate.

1.5. Research Questions

RQ1 - what are the types and stages of cataract?

RQ2 - what kind of data do we need from the cataract patients and ophthalmologists to detect the presence of the cataract?

RQ3 - how to identify cataract using machine learning algorithms and image processing?

1.6. Research Aim

Make Cataract diagnosis more accessible to the people in rural areas of Sri Lanka.

Despite the rapid advancement of new technology, visual impairment continues to be a significant problem for worldwide healthcare systems. A cataract is one of these issues cataract has become an important cause of blindness and severe visual impairment. Because smartphones are used by so many individuals in most urban areas, they can make self-diagnosing cataracts easier. The World Health Report estimated that there were 20 million people with bilateral blindness (i.e., with the eyesight of less than 3/60 in the better eye). That number will have increased to 40 million by the year 2020 (Brian and Taylor, 2001). Currently, the methods available for the detection of cataracts are based on the use of both a fundus camera or a Digital Single-Lens Reflex (DSLR) camera; both are very expensive (Pathak and Kumar, 2016). The lack of ophthalmologists and slit lamp cameras in poor and rural areas are the main causes of late cataract diagnoses. This process is time-consuming and costly for poor people as well. Early diagnosis of cataracts is necessary to preserve vision and prevent the increase in blindness. Considering all these reasons, it was decided to investigate on automated cataract grading using mobile phone images.

1.7. Project Scope

Considering the objectives and research related to the project, the scope of the project is defined as follows. The project intends on helping people in rural areas who don't have access to medical professionals' help identify if they have Cataracts in their eyes or not. The focus is to identify stages of Cataracts in human eyes by using a smartphone camera. This can be achieved by using different kinds of datasets.

1.7.1. In-Scope

The scope that will be focused on the project is as follows:

- People in the age range above 40 years are considered - The possibility of age-related cataracts in people older than 40 years is considerably high compared to people in the age range below 40 years.

- Lifestyle factors that have a significant impact on cataracts are taken into account. - Lifestyle factors that have a considerable effect on cataracts such as exposure to UV rays are taken into consideration.

- User-friendly GUI to facilitate requirement - A user-friendly GUI with essential features which will have the functionalities such as uploading an image, Analyzing the said image to predict cataracts levels and finally recommend the user to take the necessary actions according to the prediction.

1.7.2. Out-Scope

The factors that are excluded from the project are as follows:

- Prediction of cataracts in children is not included in the project - Due to the lower risk of cataracts in children, which is 1 to 3 per 10,000 children, the prediction of cataracts in children is offered as a future enhancement.

- People in foreign countries are not considered for the project - Only Sri Lankan people are considered for this project as health issues like cataracts vary from different geological locations to ethnicities differently.

1.8. Rich Picture Diagram

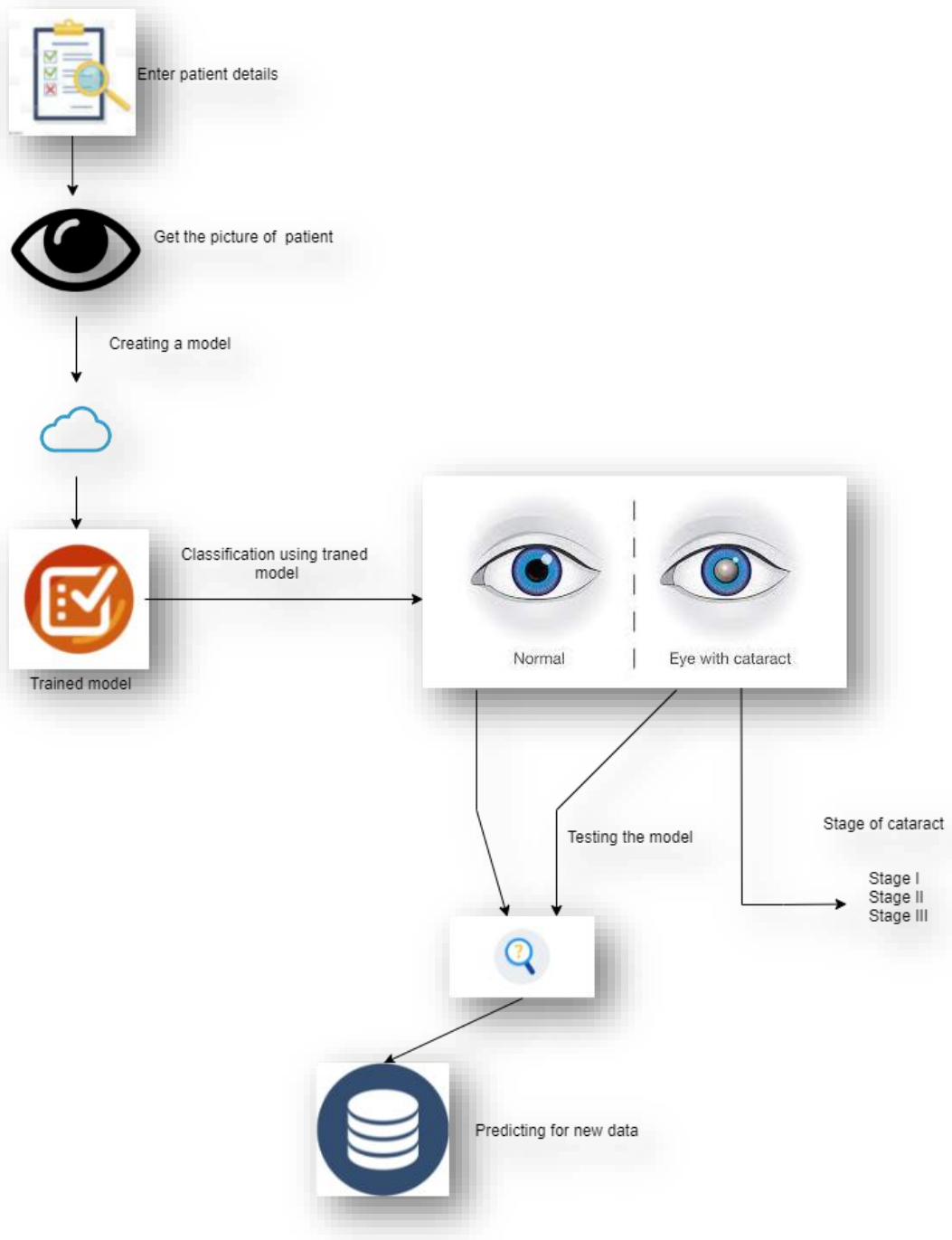


Figure 2 : Rich Picture Diagram

1.9. Objectives

1.9.1. Research Objectives

	Objective	Description
OBJ_01	Do research on Cataracts patterns by talking with medical professionals.	Discuss with a medical professional and compare the dataset with their knowledge and fix inconsistencies.
OBJ_02	Improve the accuracy of the product.	Figure out a solution to increasing the accuracy of the product.
OBJ_03	Do more research about the machine learning models	Carry out in-depth research into relevant machine learning models to make the product more accurate.

Table 4 : Research Objectives

1.9.2. Academic Objectives

	Objective	Description
OBJ_01	Finding public medical research on Cataracts patterns in human eyes.	Carry out deep research regarding people with cataracts in Sri Lanka and gather information about their ages and their Cataracts patterns.
OBJ_02	Select appropriate research approaches to create a project plan.	Analyze and choose the most appropriate research techniques and software development methodologies for an effective system build.

Table 5 : Academic Objectives

1.9.3. Operational Objectives

	Objective	Description
OBJ_01	Identify the requirements.	Gather all the functional and non-functional requirements.
OBJ_02	Implement the system.	Choose the right programming approaches and put the Cataract detection system in place. To develop the system, proper development procedures (training the model) should be followed.
OBJ_03	Identify the proper tools and algorithms.	Analyze and evaluate the algorithms, then choose appropriate tools to put the system in place.
OBJ_04	Identify architectural design requirements.	Evaluate the system and come up with a design specification that is appropriate.
OBJ_05	Test and evaluate the system.	Evaluate the system using appropriate test cases to ensure that it meets all requirements and achieves its single purpose with greater precision.

Table 6 : Operational Objectives

1.10. Resource Requirements

1.10.1. Hardware Requirements

- A smartphone with a high-definition camera for capturing data/Images
- A powerful Windows Computer with following Specifications:
    1. CPU - Intel Core i7-4790 or higher
    2. GPU Nvidia GeForce GTX 1060 4GB or higher
    3. Ram - 8GB or More
    4. Storage with a 128GB Solid State Drive (Optional)

1.10.2. Software Requirements

Languages	
Python	For the Machine Learning Backend
IDEs and Software's	
Pip	Python package management system
Jupyter Notebook	For backend programming while testing
PyCharm	For python programming
Slack	For Communication Purpose/File Sharing
Google Meet	To conduct general Meetings
Adobe XD	For creating design prototypes and wireframes
Microsoft Word	For making reports and documentation
LabelMe	For Labelling Data Sets
Google Drive	For collaborating and file sharing between team members
StarUML, Visual Paradigm	For making UML Diagrams
Omniplan	For making Gantt Charts
Git	For version controlling
GitHub	As a remote repository and to collaborate with team members
Libraries and APIs	
TensorFlow	Machine Learning Library for Image recognition/object detection
Keras	Deep Learning API for Python
CNN	Machine Learning Algorithm
NumPy	Library For Addition of Large Multidimensional arrays and matrices

Table 7 : Software Requirements

1.10.3. Data Requirements

Data requirements Since the project is under the machine learning domain. A dataset is a must have requirement. The dataset for this project should be an image dataset. And the images need to be labeled to be used for training models. The most important aspect when refining the dataset should be the clarity and the uniqueness of the data in the dataset.

1.11. Chapter Summary

This chapter provided an overview of the proposed prototype Cataract Analysis system. It contains a detailed description of the problem area and information about the project background, this chapter provided an overview of existing solutions and described the resource needs identified in the project. The next chapter provides an important overview of existing research, its weaknesses and strengths, and the rationale for the proposed solution.

Chapter 02 – Literature Review

2.1. Chapter Overview

The previous chapter went over the introduction of the Cataract Stage Analysis and in this chapter Literature review will critically evaluate on the problem domain and the existing systems and the tools and technologies available and understand clearly on the pros and cons of the project and find the best approach to implement.

2.2. Literature Review of the Domain

2.2.1. Introduction to Cataract

The most prevalent cause of reversible blindness and visual impairment in the world is cataract. Cataract blindness is more common in low-income and developing-country populations than in developed-country populations. Cataract surgery is the only treatment strategy. The findings of many medical studies are still awaited, the most pressing issue is the growing cataract blindness patients in underdeveloped countries due to a scarcity of cheap surgery and lack of knowledge. Efforts to train more cataract surgeons in these countries are falling short compared to the increased demand brought on by an ever-aging population. It is important to focus efforts and resources on designing a system for efficient delivery and diagnosis of cataract services in low-income regions to ensure countermeasures can be taken to prevent or delay cataract formation. (Lam et al 2015)

2.2.2. Causes of Cataract

Over the years, several risk factors for lens opacification have been found, including ultraviolet light exposure, Diabetes and increase in age. Ageing Age is by far the most significant risk factor for cataract, and it's commonly falsely considered that cataract is merely an accelerated version of the ageing process. This is not the case, as the change in the lens caused by aging and cataract are separate. The characteristic of age-related nuclear (ARN) cataract is oxidation. The oxidation of methionine residues and the reduction of protein sulfhydryl groups progress as the cataract advances. (Truscott, 2005). Diabetes A study done shows there is an increased development of Cataract formation in people with diabetes mellitus (DM) and patients with DM are reported to be up to five times more likely to develop cataract at an early stage (Saxena, Mitchell and Rochtchina ,2004). The way by which cataract forms could happen by several different ways one way is polyol pathway where an enzyme known as aldose reductase (AR) speeds up the breakdown of glucose into sorbitol, where the increased collection of intracellular sorbitol leads to a hyperosmotic effect causing the hydropic lens to breakdown and form cataract. (Kiziltoprak, Hasan et al 2019). Another system would be Osmotic stress where the increased swelling of the cortical lens fibers, this also happens due to the accumulation of sorbitol which results in a force added to the endoplasmic reticulum (ER) (Kiziltoprak, Hasan et al 2019)

2.2.3. Diagnosis of Cataract

Cataracts are a common cause of vision impairment in the elderly that are often noticed by patients in the early stages, and surgery is often effective in restoring vision (Foster A., 2001). However, cataract surgery remains a major medical expense in Europe and other countries. In Western countries, during the aging process, the dispersion of light does not increase uniformly as the layers of the lens get more apparent, but certain layers disperse more gently than others. As a result, each decade of human life has its own lens light scattering characteristic. A further increase in the scattering of light in the lens of the eye leads to the development of cataracts. (Prokofyeva, Wegener and Zrenner, 2012).

To identify cataracts, the doctor will review the patient's medical history and symptoms and perform an eye exam. And your doctor may perform several tests, including:

- A low vision test which uses a vision chart to measure how well a patient can read a series of letters. The eyes are tested individually while the other eye is covered.
- Inspection with a slit lamp. This allows the optician to examine the front of the eye in more detail. A slit lamp microscope is so named because it illuminates the patient's cornea, iris, lens, and the area between the iris and the cornea with a bright line of light termed a slit. Doctors can see these structures in minute parts thanks to the slits, making it easy to spot even the tiniest irregularities.
- A retinal examination in which an ophthalmologist puts a drop in the patient's eye to open (widen) the pupil. This makes it easier to examine the fundus (retina). An ophthalmologist can use a special device called a slit lamp or ophthalmoscope to inspect the lens for signs of cataract.
- Ocular tonometry. This test measures the hydraulic pressure in the eye. There are several devices available to test this. (Mayoclinic.org, 2018)

2.2.4. Cataract Related Treatments

People with cataracts can improve their eyesight with eyeglasses, magnifying lenses, or brighter lighting. However, surgery is the only way to treat a cataract. It's crucial to assess how terrible your vision is against the minor risk of surgery and the possibility that it will improve your vision before proceeding with surgery. During cataract surgery, the clouded lens is removed and replaced with a plastic lens that is introduced into the eye during the procedure. Cataract surgery improves vision in most patients, but not all. The procedure can result in posterior capsule opacification, which causes the area of the eye behind the artificial lens to become clouded. Laser surgery can be a remedy for this procedure (Harvard Health, 2014)

2.3. Literature review of Existing Systems

2.3.1. Automated nuclear cataract detection algorithm using ocular images captured by smartphone-based slit-lamp

The objective of the algorithm is to automatically detect the severity of the cataract based on the photometric appearance of the nuclear region of the lens of the eye. YOLOv3 was used to locate the nuclear area of the ocular lens. Next, the gray conjugate features of the nuclear region were assessed using a combination of a deep learning network, Shuffle Net, and a Support Vector Machine (SVM) classifier to assess the severity of the cataract. 819 previous eye images acquired with a smartphone slit lamp were used to assess the performance of the algorithm using the trained algorithm. With a Kappa of 95.4% and an F1 of 92.3%, the accuracy was 93.5%. (Hu et al., 2020). The AUC for this study was 0.9198. The proposed validation approach may determine cataract severity in 29 milliseconds and complete the categorization procedure in under a minute. This research has the potential to increase the accuracy of the examination, lower the rate of misdiagnosis, and make the medical examination less burdensome. The rating system can help non-ophthalmologists enhance the quality of their photographs. The method is especially well suited for cataract screening in poor or underserved locations with limited ophthalmic resources. It may also make ophthalmic care more accessible. Cataract detection procedure reduces the complexity of artificial intelligence detection, improves accuracy, lowers the rate of misdiagnosis, and dramatically improves medical treatment accessibility. So far, 94,246 persons have been tested in the province of China. This strategy is currently being used in Guizhou Province's Pingtang and Luodian Counties in China's anti-poverty medical activity. This approach is also employed at the international level to diagnose cataracts in Nigeria, Africa, where 1079 persons have been investigated so far. (Hu et al., 2020)**.**

2.3.2. Automated Classification of Normal, Cataract and Post Cataract Optical Eye Images using SVM Classifier

The SVM classifier was employed in this system. Vapnik et al. created a very good, supervised classifier. Reduce the empirical training error as much as possible. By maximizing the margin between the separator hyperplanes and the data, MVS aims to reduce an upper bound on the generalization error. SVMs have been utilized successfully in a variety of applications, including classification, regression, and so on (Niya C P, 2014). The roles within the three instructions of optical eye photos, including normal, cataract, and post-cataract pictures, are discovered using image processing algorithms in this work. The Big Ring Area (BRA), Small Ring Area (SRA), Edge Pixel Count (EPC), and Object Perimeter are retrieved from the optical eye vision. The functions have been statistically examined and shown to be significant in the automated category. The automated classifier then employs the same functions, as well as Support Vector Machines (SVM) for the automated category. With 94% sensitivity and 93.75 % specificity, the implications are clinically significant. The category price is sort of 90% (Nayak, 2013). The image properties of the optical eye, such as the large ring area, the tiny ring area, the number of pixels on the edge, and the perimeter of the object, are principally extracted by this system. The characteristics are statistically evaluated and determined to be significant for automatic classification. In all three types of pictures, the inner surface of the cornea is different. When comparing normal and post-

cataract photos, the inner surface of the cornea is whiter in cataract photographs. It serves as the foundation for estimating the area of the little ring. The tint of the cornea's outer surface varies between the three groups. In cataract images, the outer surface of corneal images is light compared to normal and post-cataract images. Find a large ring area. Then EPC (Edge Pixel Count) is calculated using the Cannys edge detection method. The number of white pixels within the output of the brink detection is counted using the EPC algorithm. The count in a regular photograph is only a few, but in a post-cataract photograph, it's significantly more than the usual, indicating that the count inside the cataract photograph is very high. The item perimeter is the next function extracted, for which erosion is conducted. The grey levels in the regular, cataract, and post-cataract images contain much too many unexpected changes. As a result, those images could have a lot of edges, which could be employed as a function to mirror normalcy and abnormality in the attention images. As a result, visible erosion produces an amazing outcome (Niya C P, 2014). These classifiers can also be used to assess the success of cataract surgeries using post-cataract photos. These tools, on the other hand, rarely provide 100 percent accurate results. The precision of these tools is determined by several criteria, including the size and quality of the training set, the rigor with which the training was delivered, and the parameters used to represent the input (Nayak, 2013). The downside is that classification of normal and post-cataract images is less accurate than classification of cataract images.

2.3.3. e-Paarvai: AI Based Android Mobile App to Detect Cataract

The Tamil Nadu eGovernance Agency (TNeGA) has created e-Paarvai, an intelligent smartphone application that uses artificial intelligence to detect the existence of cataracts in the eye of the person being screened. ("vikaspedia Domains," n.d.). Several AI systems for automated cataract identification and grading have also been created, based on either machine learning or deep learning methodologies. Apart from the differences in the basic architecture of AI algorithms, these previously established algorithms also differed in terms of the types of input photos employed (eg, either based on slit lamp photographs or color fundus photographs) (Goh et al., 2020). The approach is based on computer vision technology, and the AI model is implemented via an Android app that allows field personnel to record the patient's eye with their smartphone. The model then determines if the eye has a mature cataract, an immature cataract, an intraocular lens (IOL), or no cataract. With doctors already overworked, an analytics-only approach would have added to their workload and caused the entire screening process to be delayed. As a result, AI was selected as a critical component of the solution. Currently, the AI model has a 91% accuracy rate. ("vikaspedia Domains," n.d.). The development e-Paarvai's was not without its difficulties. Finding such a dataset was the most difficult challenge because the e-Paarvai software was built on free-hand eye photos collected from a smartphone. As a result, the entire training dataset must be rebuilt from the ground up. Furthermore, the TAM-DEF (Transparency & Audit, Misuse Protection, Digital Divide & Data Deficit, Ethics, Fairness & Equity) governance structure (Tamil Nadu AI Policy, 2020) is followed by the system. ("vikaspedia Domains," n.d.). The Tamil Nadu State Control Blindness Society is now using the e-Paarvai app (TNSBCS). Since its start in February 2021, the system has been expanded to 14 districts and has screened over 1280 patients in Tamil Nadu's rural areas.

2.3.4. Automatic Cataract Detection and Grading Using Deep Convolutional Neural Network

This work is supported by Beijing Natural Science Foundation (4152007), China National Key Technology Research and Development Program projects with no. 2013BAH05F02 and 2015BAH13F01 (Zhang et al., 2017). This challenge was to investigate the overall performance and performance by using a Deep Convolutional Neural Network (DCNN) to automatically detect and grade cataract. It also visualizes several the feature maps at the pool5 layer with their high-order empirical semantic meaning, providing evidence to the function illustration extracted with the aid of DCNN. The suggested DCNN type device was tested on numerous population-based, completely scientific retinal fundus photos collected from a hospital, totaling 5620 images achieved an accuracy of 93.52%, but this work failed to manage the vanishing gradient problem (Hossain et al., 2020). The following are some of the paper's recommendations: The first is that the interference of local choppy lighting and eye reflection has been eliminated by employing retinal fundus pictures after G-filter, which contributes significantly to the DCNN type. The second is that when the number of available samples grows, the DCNN type accuracies grow, and the fluctuation variety of accuracies becomes more stable. Our approach attained a satisfactory accuracy of 93.52 percent in cataract identification and 86.69 percent in assessing responsibilities independently. The DCNN classifier outperforms the newest within the performance range on this project. (Zhang et al., 2017). Using supervised learning, the suggested Deep Convolutional Neural Network (DCNN) technique can obtain file breaking results within the demanding cataract detection and grading responsibilities. In terms of accuracy and efficiency, our method outperforms the competition. The use of retinal fundus pictures following G-filter, which makes a widespread contribution to DCNN classification, was able to overcome the interference of surrounding choppy illumination and eye reflection. Discriminative functions that represent high-degree statistics are successfully and automatically retrieved using DCNN, rather than artificially Furthermore, the characteristic extraction and classifier were combined, indicating a higher level of intelligence. This method has been shown to be extremely useful in the early detection and identification of cataracts, as well as its capacity to be applied to a wide range of eye diseases.

2.3.5. A method to evaluate blurriness for cataract diagnosis in retinal images with vitreous opacity

Li et al. (Xiong et al., 2017) developed a cataract detection system based on a 2D Gaussian filter and decision tree algorithm. There are three types of capabilities extracted: pixel-wide variety of visible structures, recommend assessment between vessels and backdrop, and nearby favored deviation. A morphological strategy is presented to discover and remove such lesions from retinal visible structure segmentation to avoid false recognition of vitreous opacity as retinal structures. A decision tree is trained to categorize retinal pictures into 5 levels of blurriness based on the retrieved capabilities. 1355 scientific retinal pictures were used to test the proposed approach. Preprocessing for image-size normalization, image enhancement, function extraction, and cataract grading are the processes in the proposed automated technique for cataract grading. Image-size normalization based entirely on the diameter of the received region of interest (ROI) is completed in preprocessing to system retinal images with remarkable resolutions. After preprocessing, images are also more appealing using 2D reverse Gaussian-matched clear out, and three types of capabilities are extracted. Before the very last grading, a decision tree is trained to categorize each adjoining grade into one magnificence. In a retinal image, pathological abnormalities may emerge

and these pathological changes will impact correct feature extraction. One form of pathological alteration that is typically seen in retinal imaging of cataract patients is vitreous opacity. In comparison to previous study on cataract grading using a retinal image, this is the first time that pathogenic alterations are detected and removed to increase grading accuracy. This approach can achieve 81.1 percent accuracy in accurate cataract grading and 92.8 percent accuracy in two-class screening using many clinical pictures as test data (no cataract and cataract) (Xiong et al., 2017). In the five-grade grading, the Kappa value is 0.7435, and the variance and value are both less than 0.001, indicating that there are no significant differences between this grading and manual grading.

2.3.6. Mobile Cataract Detection using Optimal Combination of Statistical Texture Analysis

Fuadah et al study the best combination of statistical texture features for cataract identification in this project. They use K-Nearest Neighbor (k-NN) as a classification approach in this study, which will be implemented on an Android smartphone. Their findings reveal that dissimilarity, contrast, and uniformity are the best texture features to combine. The system's highest level of accuracy is 97.5 %. The system is installed on a smartphone. (Fuadah et al., 2015). The findings indicate that the proposed approaches are adequate for classifying the two types of eye conditions: normal and cataract. It's lightweight and simple to use. The relevant texture features and k value of K-NN that performs high accuracy of the system, the algorithm approach implemented on an android smartphone for creating a mobile cataract application The user interface is how the user and the system interact. The main activity and start activity of the MCataract application provide capabilities such as taking a picture or loading from the gallery, cropping to acquire the pupil area as RoI, displaying the pupil area obtained, and diagnosing the state of the eye image. (Fuadah et al., 2015).

2.3.7. Automatic Detection and Grading of Cataract Using Ultrasonic Nakagami Images

In the field of ophthalmology, ultrasound is a common diagnostic tool. For cataract diagnosis, ultrasound A scan signals from porcine lenses are obtained using an ultrasound scanner with 30–60 MHz ultrasonic transducers. B-scan and Nakagami pictures are created using acoustic characteristics such as velocity, attenuation, and backscattering signals. (Hu et al., 2020). Several research have investigated using ultrasound Nakagami imaging to objectively assess cataracts in recent years. P. Tsui et colleagues. investigated whether high-frequency ultrasonic Nakagami imaging might be used to characterize the cataract lens in vitro. (Hu et al., 2020). Caixinha and colleagues present their findings in vitro. Healthy and cataractous porcine lenses were studied using ultrasound A-scan signals. The obtained signals were used to rebuild B-mode images. The B-mode images were then used to create the parametric Nakagami images. The core portion of the lens was used to gather acoustic and spectral parameters. The B-scan and Nakagami pictures were used to obtain image textural properties. A total of 97 parameters were retrieved from 75 healthy lenses and 135 cataractous lenses. Incipient and advanced cataract lenses were divided into two groups, with 60 and 120 minutes of immersion time in a cataract induction solution, respectively. Principal Component Analysis was used to choose features from the retrieved parameters (PCA) (Caixinha et al., 2014). Later, they introduced a new multi-class SVM classifier-based technique for objective cataract classification [31]. They collected a total of 97 features from the eye lens, including acoustical and spectral features. For feature selection, principal component analysis (PCA) was performed. (Caxinha et al., 2015). Multi-class support vector machines (SVM) are applied for classification of cataract lens as incipient or advanced cataract and achieved an accuracy of up to 90.62%.

2.4. Literature Review of Tools and Techniques

2.4.1. Algorithm Selection

An algorithm is simply a set of instructions for solving a problem. So, in a simpler way it's how computers to conduct actions. Modern computers are not just following instructions. They can learn from experiences and improve on their own. This is done using machine learning. Much like human learning, machine learning also can formulate answers to questions by studying a dataset. There are two ways this can be done. (Cohen, 2021)

- Supervised learning
- Unsupervised learning

Supervised learning: Supervised learning is simply a training method to create machine learning models. The specialty of this model is that it uses specialized and labeled datasets to train the model. This is done by feeding the dataset as the input and training the model to identify the similarities for future inputs. Here are some of the most popular supervised learning algorithms out there. (IBM Education, 2021)

- Linear regression
- Logistic regression
- Artificial Neural networks
- Linear discriminant analysis
- Decision trees
- Similarity learning
- Bayesian logic
- Support Vector Machine (SVM)
- Random forests

Unsupervised learning: The main difference between supervised learning and unsupervised learning is that unsupervised learning doesn't require the dataset to be labeled. They do this by discovering hidden patterns and grouping those data without human interactions. This is called clustering. This gives this method an advantage in image recognition because it doesn't need the dataset to be pre labeled. And since that process can take a lot of time. This approach could save researchers a lot of time as well. Here are some clustering types in machine learning. (IBM Education, 2021)

- Hierarchical clustering
- K-means clustering
- K-NN (k nearest neighbors)
- Principal Component Analysis
- Singular Value Decomposition
- Independent Component Analysis

The system we are trying to implement is image recognition based. So, we are planning to use a convolutional neural network for our project. R-CNN and YOLO are the candidates at this moment. We would like to train multiple models and then pick the most accurate one at the development stage. (Johnson, 2021).

2.4.2. Technology Selection

Computer Vision is a subfield of Artificial Intelligence and machine learning which specializes in extracting meaningful data from raw images and videos. With the modern-day technologies like neural networks and machine learning algorithms the ability for computers to understand information has become higher. This is done by analyzing and training lots of data to create models over and over until they can distinct and recognize certain features in an image or a video (Marr, 2019). But to use computer vision, A platform needs to be built. And for that a programming language needs to be picked. Now Python is the most robust language for machine learning purposes. And it's because Python can work with a lot of machine learning libraries. Here are some of the libraries and resources we are considering.

TensorFlow: TensorFlow is basically an open-source machine learning library for numerical computation using neural networks. It also can be used to build deep learning models. We are planning to utilize TensorFlow for image recognition and object detection. Since we won't be using pre-trained models, we are planning to train our own models with separate datasets. (Toydemir, 2021)

Keras: Keras is a high-level deep learning API written in Python. It's importance to us is that it can run on the TensorFlow framework. Keras was developed by Google as an open-source library to support backend neural networks. (Simplilearn, 2021)

Python: The reason we are going with python is that a lot of libraries we use are built on C++.But python gives us the ability to combine all of them together and import them as python modules. A pure C++ program would be much faster, but we are going with python because of the ease of use and the complexity of the code.

CNN: CNN is also known as Convolutional Neural Networks is one of the most popular machine learning algorithms out there. The reason we are considering this is that this algorithm is way ahead at pattern recognition than other algorithms. And our cataract detection system is mostly pattern based. So, this is the reason we are considering this algorithm. (Mandal, 2021)

2.5. Chapter Summary

This chapter briefly explains the critical analysis on existing technologies, it also discusses about the current trends in Image processing Algorithms and Machine Learning methodologies. This will also emphasize on the different approaches; the past research has taken and how those approaches could be improved with the modern-day technologies. From the research that has been conducted under this chapter, it concludes that the diagnosis of cataract could be much efficient and much more accessible to people in low-income areas.

Chapter 03 – Methodology

3.1. Chapter Overview

The previous chapter was basically focusing on the literature aspect of the project. Throughout this chapter the most suitable and the efficiency of the project management methodology will be examined. This chapter discusses in detail mainly about the research approach, the design approaches and the evaluation methods that are selected to use for this project and the points that have been made to select those methods. The most suitable and the appropriate methods are also discussed in this chapter. The project risks & mitigations are also discussed. The work breakdown structure and the timeline according to the activity schedule is also discussed in this chapter. Timeline and the project deliverables are discussed in this chapter.

3.2. Research Methodology

3.2.1. Research Approach

The procedure of the study should be decided based on the beginning of the study circle. Research approaches can be further classified: deductive and inductive approaches. This is because if the research starts based on an existing hypothesis or theory and then extract more data, it will be distinguished as a deductive approach.

Figure 3 : Deductive Approach

However, the inductive approach is interested in generating new theories based on the data collected.

Figure 4 : Inductive Approach

According to the above project idea which is predicting the stages of cataract using smart phone image where this theory has been existing already the deductive approach method will be more suitable for the research approach because it focuses on testing and analyzing the data.

3.2.2. Data Collection Approach

Because sensitive data is needed for more accurate forecasts, both quantitative and qualitative data were utilized in this example. The mixing method was chosen among mono-method, multi-method, and mixing method because the analysis is focused on both qualitative and quantitative data. In this way, the main necessary quantitative and statistical data were collected through surveys and interviews. Therefore, a mixed method of research was an ideal option for successful data collection.

Required Data will be collected through various ways. They include Journal articles, online Questionnaire, Interviews with domain experts, survey data, Domain related research papers and Organizational reports.

3.3. Development Methodology

The software development life cycle (SDLC) has a set of chapters which help people to understand how software is made. How is software understood and developed? A good software engineer must be able to decide the SDLC model to be used based on project conditions, to ensure its success.

SDLC models can be considered as the most effective delivery tools for software projects. Knowing and understanding when each model should be used, as well as the pros and cons of each, is crucial in deciding which one is the best for the project.

Types of Software developing life cycles

- Waterfall Model
- V-Shaped Model
- Evolutionary Prototyping Model
- Spiral Method (SDM)
- Iterative and Incremental Method
- Agile development

3.3.1. Agile Development

We selected Agile development as the software development life cycle model in this project. The reason and advantages are mentioned below, why we selected as Agile as our life cycle model.

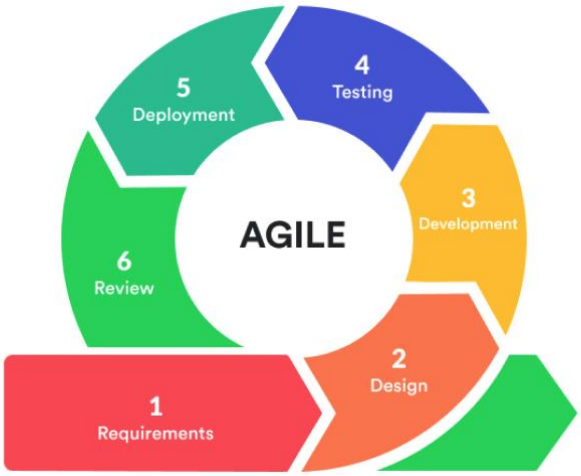


Figure 5 : Agile development life cycle model

Description:
This model is built on iterative and phased development in which needs, and solutions develop due to cross functional group corporations.

The usage:
It may be used for any type of project, but it demands greater customer participation and contact. It can also be used when a client requests that a functional need be achieved in less than three weeks, but the criteria are ambiguous. This will enable the development of more valuable and useable software items early in the process, enhancing customer satisfaction.

Advantages:

- Decrease the time it takes to use certain system features.
- There is no chance of guessing with face-to-face conversation and continuous input from the customer agent.
- Provides high quality software as well as a satisfied customer in the shortest possible time.
- We can detect project problems and changes in advance.
- Developer's initial intervention

Disadvantages:

- Heavy documentation was needed.
- Allows for a specific set of processes.
- Defines increments based on function and feature dependence.
- The line needs more customer participation than approaches.

3.4. Design Methodology

There are 2 main types of software design methods. They are object-oriented analysis and design (OOAD) and structured systems analysis and design methods (SSADM).

3.4.1. Object-oriented analysis and design (OOAD)

Object-oriented analysis and design (OOAD) is a technical approach for assessing and creating an object-oriented application, system, or enterprise. In today's software engineering, OOAD is usually done iteratively and progressively. OOAD techniques provide analysis models (for OOA) and design models (for OOD). Object-oriented analysis is distinct from other techniques of analysis in that it structures requirements around objects, which include both actions (processes) and states (data) that are patterned after real-world objects with which the system interacts. Other or traditional analytical approaches examine the two components separately: processes and data. ER diagrams can be used to represent data, while flow charts or structure charts can be used to describe actions. On the conceptual model developed during object-oriented analysis, a programmer adds Object-oriented design implementation restrictions.

3.4.2. Structured Systems Analysis & Design Method (SSADM)

The Structured Systems Analysis and Design Method is a systems-based approach to information system analysis and design.

The three most important SSADM approaches are as follows:

Logical Data Modelling:

The process of defining, simulating, and documenting the data requirements of a developing system. The result is a data model comprising entities (things about which a corporation must keep records), characteristics (information about the entities), and relationships (connections between entities) (associations between the entities).

Data Flow Modelling:

The process of determining, simulating, and documenting the flow of data through a computer system. Data Flow Modeling investigates processes (activities that modify data from one form to another), data stores (data storage locations), external entities, and data flows.

Entity Event Modelling:

Entity Behavior Modeling identifies, models, and documents the events that effect each entity, as well as the order in which they occur, whereas Event Modeling creates the mechanism to integrate entity life histories for each event.

3.4.3. Differences between structured and Object-oriented analysis.

Structured Analysis	Object-oriented analysis
They focus on process and procedures of system.	They focus on data structure and real-world objects that are important.
For various reasons such as planning, analyzing, developing, implementing, and supporting an information system, it uses the System Development Life Cycle (SDLC) approach.	It refines and extends our design using incremental or iterative technique.
It is suitable for well-defined projects with stable user requirements.	It is suitable for large projects with changing user requirements.
Risk while using this analysis technique is high and reusability is also low.	Risk while using this analysis technique is low and reusability is also high.

Table 8 : Differences between structured and Object-oriented analysis

3.4.4. Advantages of Using SSADM

They use System Development Life Cycle (SDLC):
The main reason for choosing SSADM over OOAD is that it uses the system development life cycle. SDLC is important for this project because it uses planning, analyzing, designing, implementing, and supporting the system as a cycle. And since the project is well defined and it's smaller in size this methodology fits perfectly for this project.

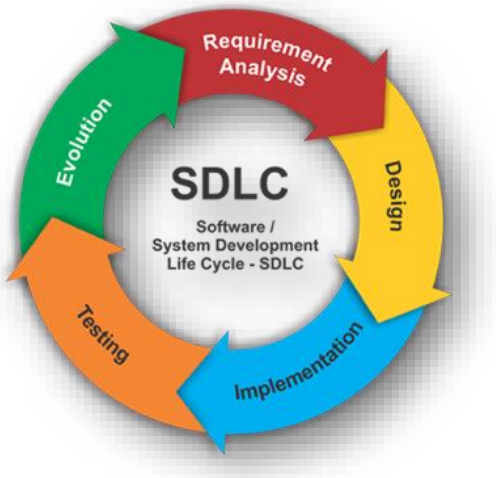


Figure 6 : Software development Life Cycle

The suitability for well-defined projects with stable user requirements:
Since the project plan is well-defined and discussed earlier, this method suits best for this project. The objectives are simple enough that they do not have to be refined with incremental development. And since the code reusability is not important for this project it is better to just plan and develop according to the target audience.

The ease of structuring with diagrams:
Another main reason for using this methodology is that it uses multiple types of diagrams like Data Flow Diagrams, Entity Relationship Diagrams and Data Dictionaries for the planning and designing phase. This is important for this project since everything needs to be documented properly for the clients (Module team) satisfaction. By doing this it also makes development easier since the project goals can be well defined.

3.4.5. Structure of SSADM

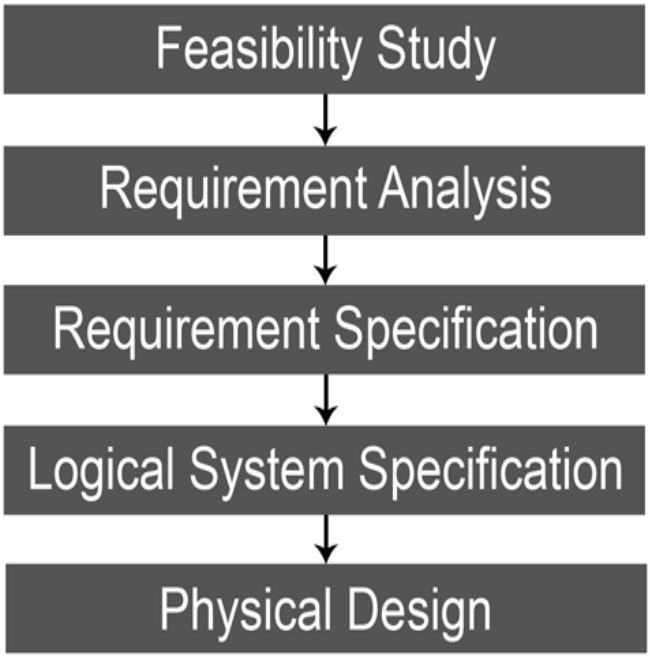


Figure 7 : Structure of SSADM

Feasibility Study: The main goal of studing the feasibility is to determine whether the product/project can acually be completed within the resources that are currently allocated to the development team. Aspects like cost, technologies used are considered in this phase.

Requirements Analysis: The goal for this phase is to investigate the similar products/projects in the market and identifing the missing parts that can be improvemed. And the next step is to develop multiple options to overcome these issues according to the requirements.

Requirements Specification: This phase emphasis on writing a detailed specification documents which contains all the details about the functions, prototyping requirements and much more about the technicalities. Diagrams like ERDs and DFDs are implemented in this stage.

Logical System Specification: The goal for this section is to design the program logic. What is meaned by this is implementing the specific tasks that the programs have to do inorder for it to be functional.

Physical Design: This section is where the user interfaces are designed. Aspects like the userbase and the ease of accesibilty are considered in this phase. And with that information the user interfaces are will be designed and developed.

3.5. Evaluation Methodology

Since there are various types of evaluation, it is necessary to adjust the evaluation method according to the content of the evaluation and the purpose of the evaluation. It is important to understand the different types of Evaluation that can be performed during the life cycle of a program. And when should they be used? The most important types of Evaluation are formative, Process, impact, and Outcome Evaluation.

Formative Evaluation

Formative evaluation is utilized prior to the design or execution of a program. It collects information on the program's requirements and creates a baseline for future monitoring. It can also point up areas of concern and give information about the program's goals. This supports project managers in defining their major problems and areas of focus, as well as creating awareness for the initiative among the selected population prior to its launch.

Process Evaluation

Process assessment examines the success of your program's methods, as well as its activities, quality, and who it serves. Its data is important for identifying inefficiencies and streamlining processes, as well as communicating the status of the program to third parties.

Outcome Evaluation

Outcome evaluation is focused on the long repercussions of the program and is frequently used to evaluate the program's purpose. Therefore, outcome evaluation determines how well the program's aim was fulfilled.

Impact Evaluation

Impact evaluation looks at the entire program from beginning to end to see if it was a success. Because it focuses on the long-term impact, impact evaluation is useful for evaluating long-term changes brought about by a system or making policy adjustments or program modifications. The goal has been accomplished.

These types of methodologies were used throughout the Research for Evaluation.

3.6. Project Management Methodology

Project management methodologies are essential to ensure that a project has a conventional structure, which can used to plan, manage and control how work is organized and completed. There are several different project methodologies each having different management structures each being optimized to better suit specific projects depending on the project requirements.

3.6.1. Considered project management methodologies

- Agile methodology
- Scrum methodology
- Waterfall methodology
- Critical path methodology

3.6.2. Summary of considered project management methodologies

Scrum- work is divided into short cycles known as sprints, work is taken from a backlog where the workload is managed, next the sprint is reviewed, and any necessary changes are made before starting the next sprint. This method has an unrealistic team cooperation and the rate at which results must be produce for a medium scale project such as the one being conducted.

Waterfall - most common methodology, project tasks are completed in a sequential manner, where each stage must be completed to move to the next stage, this project is suitable if the end goal of the project is clearly defined and the stakeholders know all the desired requirements, this method is not suitable due its inflexibility to change.

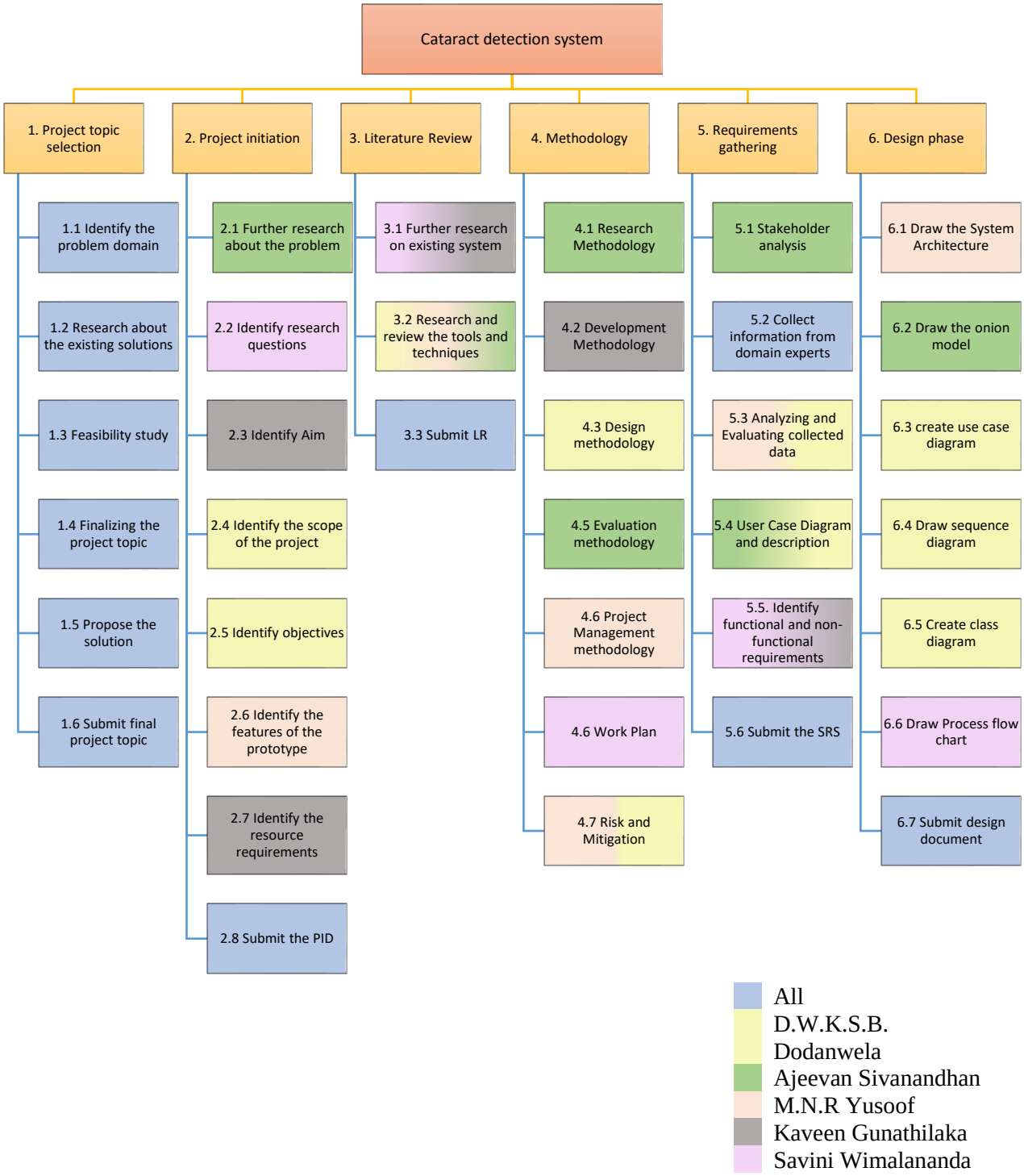
Critical path- this method consists of analyzing and scheduling all the prime concerns of the project alongside their dependencies, this method is suitable for large-scale and complex projects which have strict plans and have several dependencies. The project being conducted is not large scale and the entire project cannot be visualized to identify the most concerning tasks.

3.6.3. Selected Project management methodology

With critical analysis of available project management methodologies, agile method was selected considering the key factor of the adaptability enabling risks to be reduced as changes can be made even at the latter stages of the life cycle, other factors include not being able to visualize the outcome, the involvement of the stakeholder at every stage. Another deciding factor is the degree of collaboration where all the tasks needed to be done is recorded in a backlog and team members work through them in that stage or cycle.

3.7. Teamwork Breakdown Structure (WBS)

The reason of a work breakdown structure is to make a complicated challenge extra manageable. Breaking it down into smaller components lets for more than one group of contributors to work on it at the same time, ensuing in better team productiveness and less complicated task management. For this project, a following WBS chart was created to assist the Team in completing it.



3.8. Gantt Chart

Gantt Chart has been moved to **Appendix - A** for better understanding and Clarity.

3.9. Usage of Project Management and Collaboration Software In the project

GitHub: GitHub simply put is a software that provides Git Hosting. Git is a version control and source code management software which allows multiple users to work on different copies of the same project offline. And it also allows the ability to merge two copies of the project together much more easily. It is primarily useful when working in a team with multiple people working on the same code as it can keep track of and review everyone's individual contributions to the project. It also provides features to back up and retrieve code from any of the team members in case of an emergency where a team member's code gets corrupted. Git utilizes a branch system for merging code together, so it can track every revision in the code's history according to the date and the individual who is responsible for it.

Slack: Slack is a business communication Platform where it is used by large group organization for collaboration, and reason for using slack over other collaboration software (e.g., Discord) is because they keep text chats more organized particularly while working in teams, they have better integrations with business applications using third party bots from google drive, Zoom and Twitter and this software is built for workspace administrators while other software's mostly focus on community moderators which is similar to a public website.

Google meet: We use google meet as a video conferencing platform and Google Meet is a free service available to all Google users. Up to 100 people can participate in the free version. Furthermore, unlike Zoom, which limits free users to a 40-minute conference, the duration limit for free users is 60 minutes. It also connects with Google Docs and Google Calendar, allowing users to make calls or join meetings with a single click. Google Meet also has the advantage of being quite simple to use. It's as easy as clicking or touching a link to join a meeting. When you click a link, it will either open a web browser or an app that will take you to the meeting. One of the most significant advantages of Google Meet over Zoom and other video conferencing systems and messaging applications is that it is part of the Google Workspace suite of tools and apps, which includes Google Docs and Google Calendar, which we use to complete our project.

Google docs: Used to create project documentation and keep track of individual work, was chosen based on its collaborative features, in which every member of the project can add, remove, or alter the content of the document, and provides detailed reports of any changes made to the document, including the name of the member, date, and the section of the document which was edited. Allows each member to stay up to date on the progress of the other members and locate all documentation in one document. Microsoft Word was also considered for use, Although the document's editability and formatting were superior to Google Docs, it was not selected because of the inability to have a single document that all members could access, edit, and see changes made in real time.

Trello: Trello is an excellent web-based project management and communication tool that lets you plan all your projects in one location. It's very visible and configurable, making it easier to collaborate on a project with others, especially if they're in different physical locations. Trello gives you a quick overview of what's being worked on, who's working on it, and where something is in the process.

Proof of Project management and collaboration software has been moved to **Appendix-C** for better understanding and Clarity.

3.10. Risks and Mitigation

Risk Item	Severity	Frequency	Mitigation Plan
Dataset Integrity Since the data contains image files there is a risk of images being unrecognized due to the lack of image quality. And it could result in the machine learning model being inaccurate.	9	Low	The mitigation for this is to thoroughly analyze the dataset and consult a domain expert to authenticate the images.
Limited development knowledge Since the project is machine learning based there are complex software development concepts that require a higher level of understanding about the field.	8	Medium	Learning more about machine learning by watching tutorials online and Consulting experts in the field.
Updates and error handling While the project is in development, Additional features may need to be added or removed depending on the requirements.	6	High	Using a development methodology such as Agile to return to the previous stages of the SDLC.
The Inability to collect statistical information locally. Due to the Covid-19 pandemic situation in the country it is nearly impossible to travel into rural areas of Sri Lanka and gather information for research purposes.	7	Medium	Finding research papers online, gathering data through online communities such as Kaggle and submitting questionnaires via email.

Table 9 : Risks and Mitigation

3.11. Chapter Summary

This is a small summary for the above chapter which discusses the methodology that was used for the project. When considering the project management methodologies there are several methodologies such as Agile Methodology, Scrum methodology, Waterfall methodology, Critical path methodology In Development Methodology we use Agile Development as the software development life cycle model in this project which is built on iterative and incremental development, in which requirements and solutions evolve because of cross-functional team cooperation. The key advantage of SSADM over OOAD is that it adheres to the system development life cycle. The SDLC is critical for this project since it employs a cycle to plan, analyze, design, implement, and support the system. And, because the project is well-defined and short in scope, this methodology is ideal for this project as the Design Methodology. GitHub, slack, google meet, google docs and Trello are used for project management and collaboration software in this project.

Chapter 04 – System Requirement Specifications (SRS)

4.1. Chapter Overview

The previous chapter focused on the methodologies of project management. This chapter is a software requirements specification that primarily identifies and documents aspects and concepts that identify appropriate stakeholders, requirements for research, use cases, and functional and non-functional requirements. It focuses on the explanation of use cases, Domain model.

4.2. Stakeholder Analysis

Stakeholder investigation is the method of identifying these individuals prior to the start of a project. This grouping is based on participation, interests, and project effect. It then finds the optimal way for each of these stakeholders to be involved and communicated.

4.2.1. Onion Model

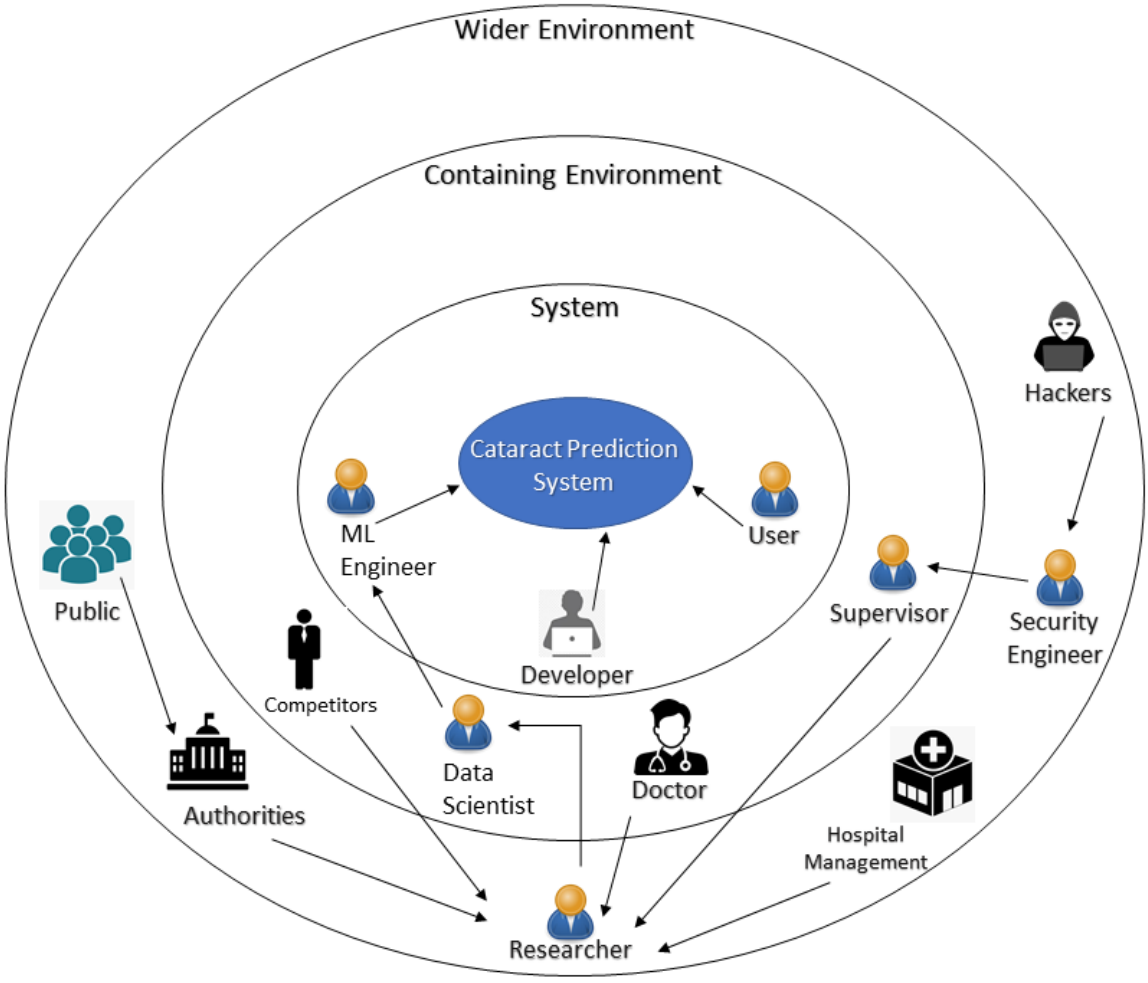


Figure 8 : Onion Model

4.2.2. Stakeholder Descriptions

Stakeholder	Viewpoint
Functional beneficiary	
User	Uses the system Implemented by the Developers Using Machine Learning Techniques
Financial beneficiary	
Hospital Management	By using the system Management could reduce costs on equipment and services.
Social beneficiary	
Hospital Management	By using the system, Hospitals will promote their Management in the Health sector
Operational beneficiary	
Developer	Developer implements Designs and tests the system
ML Engineer	Provides with necessary machine learning techniques and Algorithms to improve Prediction
Data Scientist	Helps in Preprocessing Data which is required for the model
Negative Stakeholders	
Hackers	Wanted to disturb system flow and steal confidential Data.
Competitors	Wanted to know unique system specifications and develop a better system than the existing system.

Regulatory	
Public	Provide necessary feedbacks and concerns on the system
Authorities	Check on the system frequently on the data input/output whether they are false or not
Experts	
Doctors	To get proper information and verification of data and get advice related to the problem domain
Supervisor	Overlooks and controlling the development team.
Researcher	Provides insightful data using analysis of existing systems and methods and studies the difference in tools and technologies used in the existing system
Neighboring systems	
Security Engineer	Provides computer security to protect confidential data from external entities (Hackers)

Table 10 : Stakeholder Descriptions

4.3. Selection of Requirement Elicitation Techniques/Methods

4.3.1. Brainstorming

Brainstorming is a requirement elicitation technique used to gather information and solve problems on the project; it is carried out by team members during team meetings to generate new ideas and find solutions to problems.

Each member is given an equal amount of time to express all their ideas, and this serves as a collective knowledge gathering process, with the best idea chosen at the end.

4.3.2. Observation

The most efficient way to collect data for the required system is to observe related works. Analyzing existing systems allows for the identification of patterns and trends that can be used to improve the next new system.

4.3.3. Questionnaire

Google forms were used to share questionnaire forms related to this project; this is a set of questions given to the public in order to quantify their thoughts; after collecting the responses, the data is used to narrow down and identify specific interests; the questions should be direct and easy to understand.

4.4. Discussion/ Analysis of Results

4.4.1. Brainstorming

Advantage	Disadvantage
• Promotes collective thinking and creative thinking. • Ideas and solutions to problems are found in shorter periods of time. • Promotes equal participation.	• Team members tend to argue when deciding on a final idea. • Multiple duplicate ideas may be a common occurrence.
Results	
• the project problem and solution were identified • the technology required and feasibility issues identified	

Table 11 : Analysis of requirement elicitation through Brainstorming

4.4.2. Observation

Advantage	Disadvantage
• implementation process can be easier to understand, and a proper approach can be taken • the observer can get a rough idea on what features may be of importance • limitations of current systems can be identified and overcome • Saves time	• Some areas may not be understood, since prior knowledge is required • Knowledge-based meaning a lot of research needs to be done to get a clear idea of the system
Results	
• Able to identify and add features which current cataract detection systems do not include	

Table 12 : Analysis of requirement elicitation through Observation

4.4.3. Questionnaire

Advantage	Disadvantage
• Ease of collecting data from large audience • Less time required for participants to respond as multiple people can fill the form at the same time • Questionnaire allows the anonymity to the participents.so the respondents feel more comfortable in submitting a response.	• Depending on the questions, some questions may not be clear to all participants • Limited number of responses due to not all participants filling the form • Data obtained can be unreliable as the audience is of a wide category and not one specific group.
Results	
• Able to get enough responses proving that a system for detecting cataracts in rural areas is needed.	

Table 13 : Analysis of requirement elicitation through a Questionnaire

- Questionnaire Findings and Analysis has been moved to **Appendix - B** For more clarity.

4.5. Use Case Diagrams

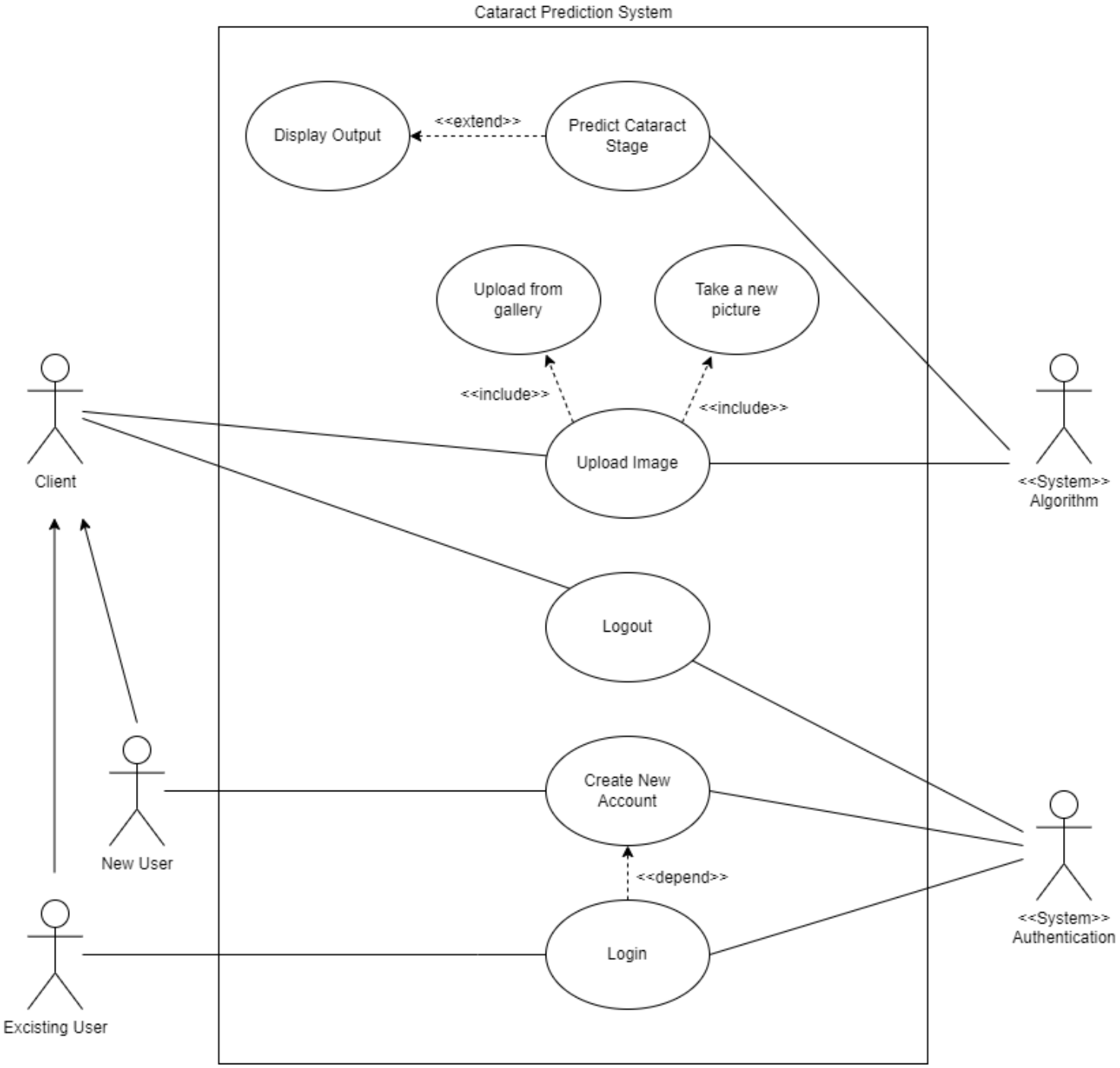


Figure 9 : Use case Diagram

4.6. Use Case Descriptions

4.6.1. Use case description for Creating new account

Use Case Name	Create new account	
Use Case ID	UC-001	
Description	Registering and creating a new account for a new user	
Priority	High	
Primary Actor	New User	
Supporting Actors	N/A	
Pre-Conditions	The user should not exist in the database The username should be unique The password should be valid	
Trigger	User selecting Create New Account	
Main flow	**Actors**	**System**
	2. Enter username 4. Enter password 6. Agrees to the user agreement	1. Prompt for email 3. Prompt for password 5. Prompt for the user agreement 7. Add user to the database
Exception flow	**Actors and System**	
	The password doesn't match the password conditions The user has declined user agreement The user is already registered in the system	
Alternate flow	**Actors**	**System**
	N/A	N/A
Post Conditions	The user is registered in the system	

Table 14 : Use case description for Creating new account

4.6.2. Use case description for Login

Use Case Name	Login
Use Case ID	UC-002
Description	Logging in an already existing user

Priority	High	
Primary Actor	Existing User	
Supporting Actors	N/A	
Pre-Conditions	The user must exist in the database	
Trigger	User selecting Login	
Main flow	**Actors**	**System**
	2. Enter username/email 4. Enter password	1. Prompt for username/email 3. Prompt for password 5. Authenticate user 6. Display the main menu
Exception flow	**Actors and System**	
	The username/email entered is not on the database The password entered does not match	
Alternate flow	**Actors**	**System**
	N/A	N/A
Post Conditions	The user is logged in to the system The main menu will be displayed	

Table 15 : Use case description for Login

4.6.3. Use case description for Logout

Use Case Name	Logout
Use Case ID	UC-003
Description	Logging out from the system
Priority	Medium
Primary Actor	Client
Supporting Actors	New User Existing User
Pre-Conditions	The user must be logged into the system
Trigger	User selecting Log Out

Main flow	**Actors**	**System**
	1. User clicks Log Out	2. Exit the program
Exception flow	**Actors and System**	
	N/A	
Alternate flow	**Actors**	**System**
	N/A	N/A
Post Conditions	N/A	

Table 16 : Use case description for Logout

4.6.4. Use case description for Uploading Image

Use Case Name	Upload Image	
Use Case ID	UC-004	
Description	Uploading an image to the system	
Priority	High	
Primary Actor	Client	
Supporting Actors	New User Existing User	
Pre-Conditions	The user must be logged into the system The user should have a valid image to upload	
Trigger	User selecting Upload Image	
Main flow	**Actors**	**System**
	1. User uploads image	2. Sends the image through the algorithm
Exception flow	**Actors and System**	
	N/A	
Alternate flow	**Actors**	**System**
	N/A	N/A
Post Conditions	N/A	

Table 17 : Use case description for Uploading Image

4.6.5. Use case description for Uploading from Gallery

Use Case Name	Upload From Gallery	
Use Case ID	UC-005	
Description	Uploading an image to the system from the user's local storage	
Priority	High	
Primary Actor	Client	
Supporting Actors	New User Existing User	
Pre-Conditions	The user must be logged into the system The user should have a valid image to upload in their local storage	
Trigger	User selecting Upload from Gallery	
Main flow	**Actors**	**System**
	1. User selects uploads image 3. Selects the image that needs to be uploaded	2. Prompt the user to open the gallery 4. Sends the image through the algorithm
Exception flow	**Actors and System**	
	N/A	
Alternate flow	**Actors**	**System**
	N/A	N/A
Post Conditions	N/A	

Table 18 : Use case description for Uploading from Gallery

4.6.6. Use case description for Taking New Picture

Use Case Name	Take a new picture
Use Case ID	UC-006
Description	Takes a new picture and uploading it to the system
Priority	High
Primary Actor	Client
Supporting Actors	New User Existing User

Pre-Conditions	The user must be logged into the system The device should have a camera built in	
Trigger	User selecting Take a new picture	
Main flow	**Actors**	**System**
	1. User selects take a new picture 3. Takes a picture of a human eye.	2. Prompt the user to open the camera and take a picture 4. Sends the image through the algorithm
Exception flow	**Actors and System**	
	N/A	
Alternate flow	**Actors**	**System**
	N/A	N/A
Post Conditions	N/A	

Table 19 : Use case description for Taking New Picture

4.6.7. Use case description for Predicting cataract stage

Use Case Name	Predicts cataract stage	
Use Case ID	UC-007	
Description	Predicts the cataract levels in an image	
Priority	High	
Primary Actor	System	
Supporting Actors	N/A	
Pre-Conditions	The user must be logged into the system A picture needs to be uploaded to the system	
Trigger	User selecting Upload Image	
Main flow	**Actors**	**System**
		1. Predicts cataract levels according to the machine learning algorithm

Exception flow	**Actors and System**	
	N/A	
Alternate flow	**Actors**	**System**
	N/A	N/A
Post Conditions	N/A	

Table 20 : Use case description for Predicting cataract stage

4.6.8. Use case description for Display Output

Use Case Name	Display Output	
Use Case ID	UC-008	
Description	Displays the prediction to the user	
Priority	High	
Primary Actor	System	
Supporting Actors	N/A	
Pre-Conditions	The user must be logged into the system The algorithm should have produced a usable prediction	
Trigger	The machine algorithms predicting the result	
Main flow	**Actors**	**System**
		1. Displays the prediction to the user
Exception flow	**Actors and System**	
	N/A	
Alternate flow	**Actors**	**System**
	N/A	N/A
Post Conditions	N/A	

Table 21 : Use case description for Display Output

4.7. Functional Requirements (with prioritization)

The process of defining the activities required by a system for it to run smoothly and in compliance with requirements is known as functional requirement analysis.

FR No	Requirements Description	Priority
1	The user should be able to create an account in the application.	High
2	The user must be able to log into the application, and the system should obtain the credentials of the user logged in to the session.	High
3	Functionality for application users to see how to use cataract detection applications.	High
4	The functionality for application users to take eye photos that will be detected.	High
5	Functionality for application users to detect cataracts.	High
6	The functionality for application users to see the results of cataract detection.	High
7	The functionality of the system to validate the object to be detected.	High
8	The functionality of the system to classify iris cataracts and normal iris.	High
9	The user should be provided with a personalized articles collection regarding their cataract type.	Medium
10	The user should be able to modify or delete his/her account.	Low

Table 22 : Functional Requirements

4.8. Non-Functional Requirements

Non-Functional Requirements	
Security	Is your product capable of storing or transmitting sensitive data? Is there a set of rules that your IT department must follow? In your industry, what are the security best practices?
Capacity	What storage requirements does your system have now and in the future? How will you scale up your system to meet rising volume demands?
Compatibility	What is the bare minimum in terms of hardware? What operating systems, and which versions of those operating systems, must be supported?
Reliability and Availability	Under typical conditions, what is the critical failure time? Is it necessary for a user to have access to something at all times of the day and night?

Performance and scalability	What is the system's response time? How much will this performance vary as workloads increase?
Maintainability + Manageability	How long does it take to repair components, and how easy is it to handle the system as an administrator? You might also define Recoverability and Serviceability under this umbrella.
Performance	How rapidly does the system react to user actions, and how long does a user have to wait for a certain process to take place?
Environmental	What types of environments will the system be expected to perform within?

Table 23 : Non-Functional Requirements

4.9. Chapter Summary

This chapter Summarizes on all System Requirement Specifications and the research concepts including Stakeholder analysis with Onion Model with the viewpoint of each and every stakeholder and discussing the Functional and Nonfunctional requirements and a brief on the use case diagrams of the model with descriptions with relevance to the domain.

Chapter 05 – Social, Legal, Ethical and Professional Issues

5.1. Chapter Overview

The chapter that follows examines the BCS code of conduct, SLEP issues, and their interrelationship. In addition, the steps taken to mitigate the SLEP issues encountered in the project are detailed in the following chapter.

5.2. Social, Legal, Ethical and Professional Issues

5.2.1. Social Issues

The project prototype was created in just English, which may have an impact on persons who do not speak English. As a future improvement, multilanguage support will be implemented.

Even though the questionnaire results were considered, sensitive data about the users was not collected to protect their privacy.

Due to the project's broader scope and a lack of experience and knowledge in that domain, the recommended that this framework be tried with any applications that send secure information until addressing its security problems.

5.2.2. Legal Issues

Throughout the development cycle of the project data privacy was given a higher priority. Since the dataset contains personal medical information. The legality for using the said data needs to be carefully analyzed. The dataset was taken from Kaggle. And the usage information for the dataset was licensed as public. And the external data that was added to the dataset as expansions were sorted and were given by a domain expert who has attained them from their clients with their approval.

For the back-end development, software libraries such as TensorFlow and Keras were utilized. And for the testing Jupyter Notebook was used. All the software's mentioned above were open source and did not require any licensing. For the software's that required licensing they were all attained through legal means

5.2.3. Ethical Issues

All the information and content collected for the research purpose and for SRS document were given proper credits to authors and the references and citations were added in all necessary places.

All the feedback collected from domain experts were from their experiences in their corresponding fields and those data were provided with the full consent of the domain expert.

While sending out surveys all the respondents had their freedom to ignore or respond to the Questionnaire. No personal and controversial data were collected, while accepting responses for the survey. (e.g.: email, age, religion, gender, race)

5.2.4. Professional Issues

Professional Conduct establishes a foundation for individual accountability and professional behavior. public interest - Based on the current state of the pandemic, domain expert guidance and surveys were ensured to be carried out via online methods such as zoom and Google forms. Punctuality - To avoid any inconvenience, meetings with mentors and additional meetings were scheduled, and all project progress was prepared ahead of time and presented to mentors on a weekly basis.

5.3. Chapter Summary

This chapter focuses on the BCS code of conduct, as well as SLEP concerns that are relevant to our research project and how they were addressed. It will highlight the SLEP policies adopted by members to get the most out of the project after discussing future system developments. Professional, Ethical, Legal, and Social are the four components that make up the ethical aspects of research analysis. The goal of enforcing them is to ensure that sensitive user data is not abused but is instead used properly by the parties involved. The Author's project does not use user data that has already been obtained and does not rely on data collection to function

Chapter 06 – System Architecture and Design

6.1. Chapter Overview

This Chapter will more likely investigate the system architectures and the design aspects of the project, The system architecture design which use the layered architecture and some System design diagrams which includes a class diagram, sequence diagram and a process flow chart with a high-fidelity prototype design for UI used for the project.

6.2. System Architecture Design

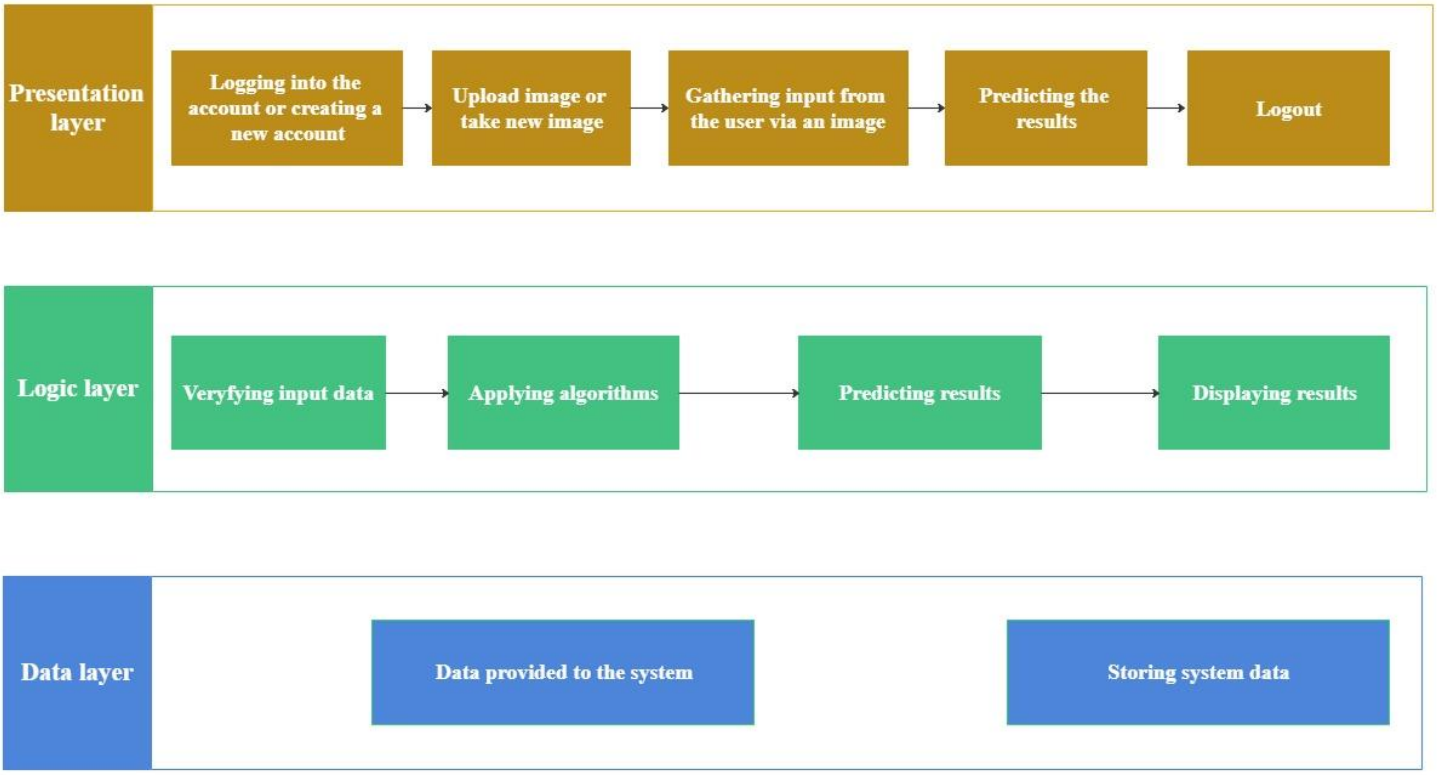


Figure 10 : System Architecture Design

Presentation Layer	
Logging into the account or creating a new account	Every user is given two options to either create a new account or login into an existing account in order to connect with the system
Gathering input from the user via an image	For the system to gather data from the user, he/she must be logged into the system and the user must have a valid image which can be uploaded using stored image in the gallery or an image taken at that instance with the mobile phone camera.
Predicting the results	after the image has been uploaded to the system, the image is sent through the algorithm and results are predicted accordingly.

Table 24 : Description for Presentation layer

Logic Layer	
Applying the algorithm	The algorithm is a crucial aspect of the project since without it the cataract prediction system won't be able function.
Processing input data	After the user uploads an image file to the system it goes through a process where the image gets resized and aligned for the cataract prediction engine to easily identify the target area.
Generating result	After the image has gone through the algorithm the results will be generated by the prediction engine. And those said results will be displayed for the user.

Table 25 : Description for Logic Layer

Data Layer	
Data provided to the system	Every user is given two options to either create a new account or login into an existing account to connect with the system
Storing system data	For the system to gather data from the user, he/she must be logged into the system and the user must have a valid image

Table 26 : Description for Data Layer

6.3. System Design

6.3.1. Class Diagram

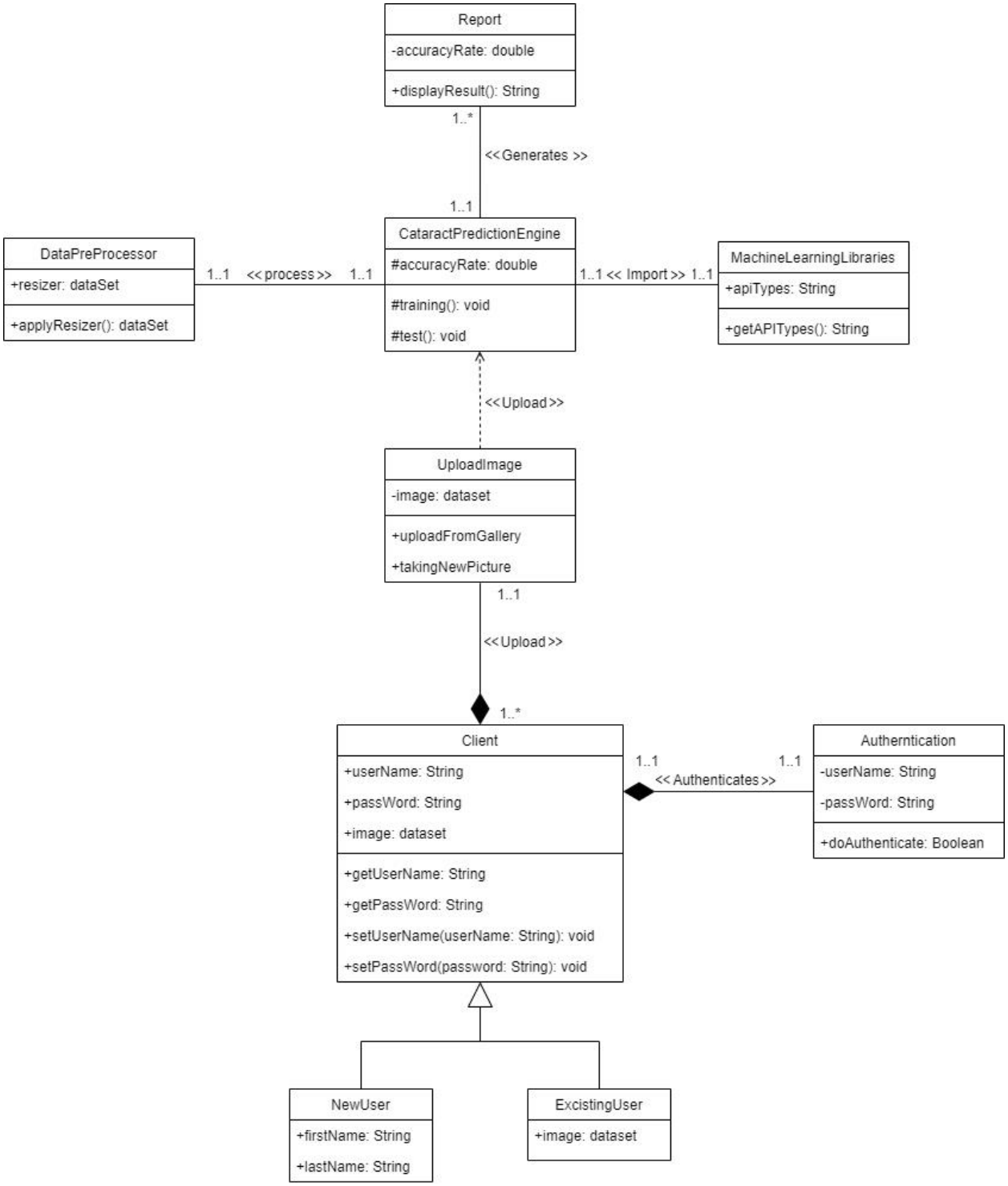


Figure 11 : Class Diagram

6.3.2. Sequence Diagram

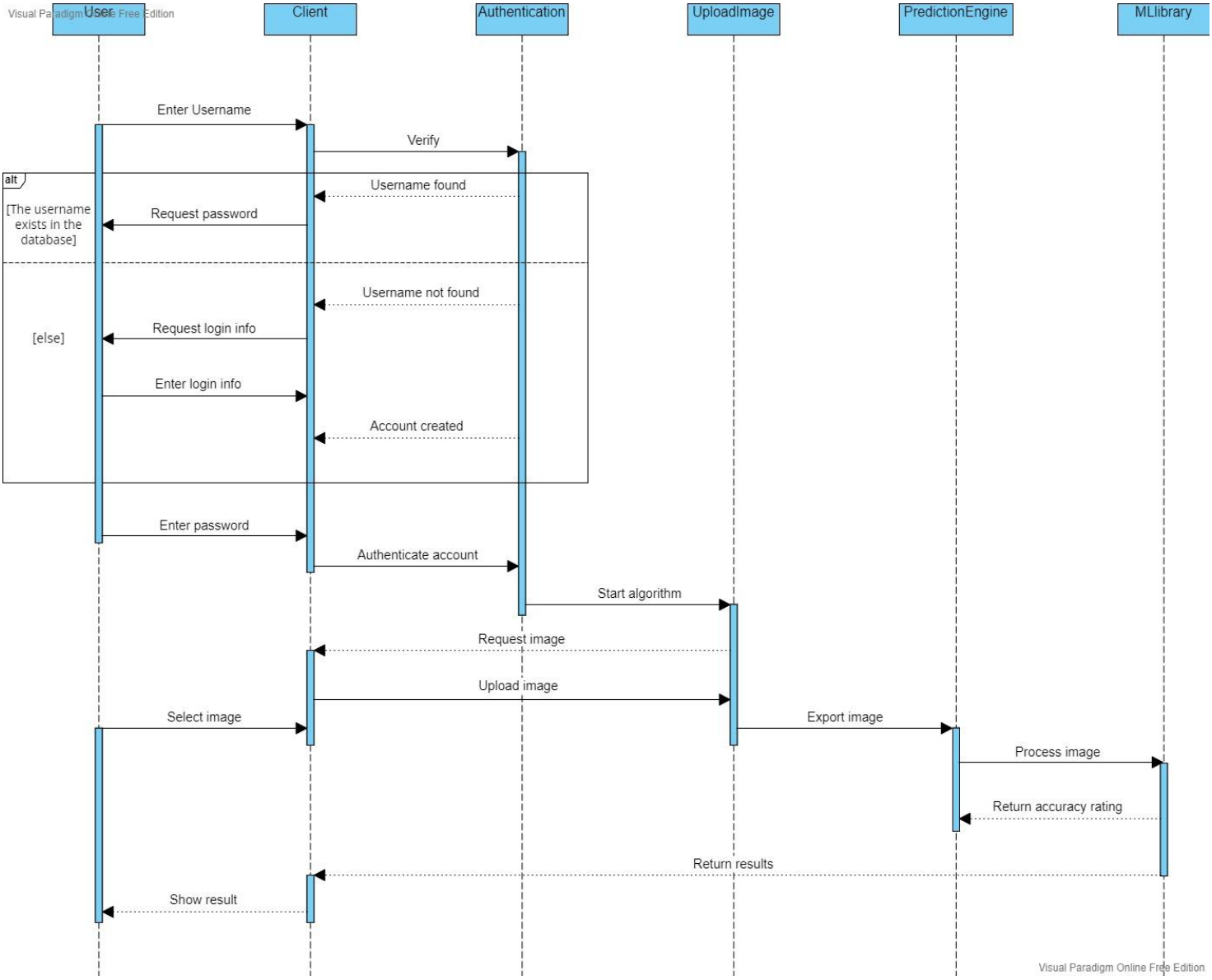


Figure 12 : Sequence Diagram

6.3.3. Process flow chart

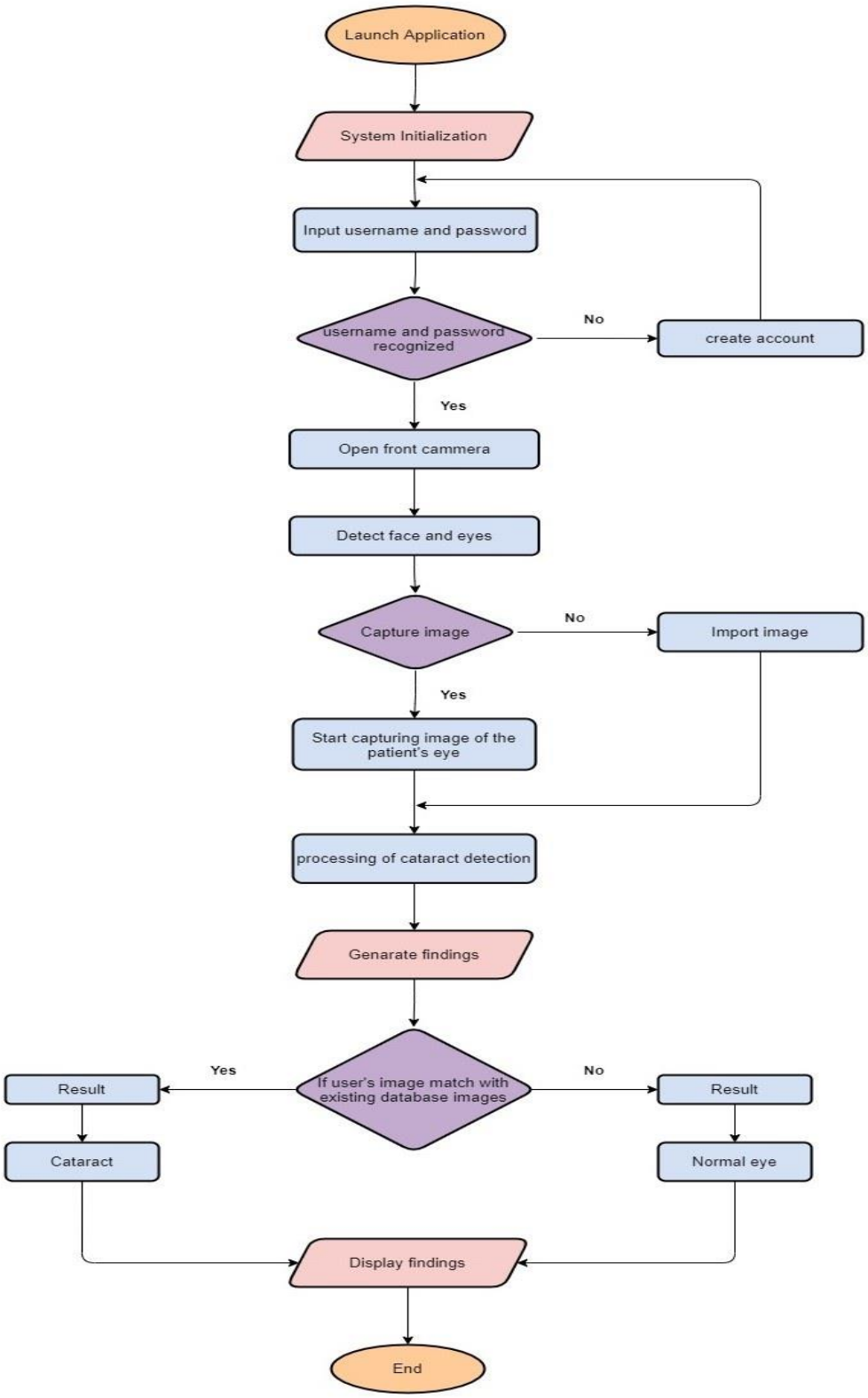


Figure 13 : Process flow chart

6.3.4. UI Design

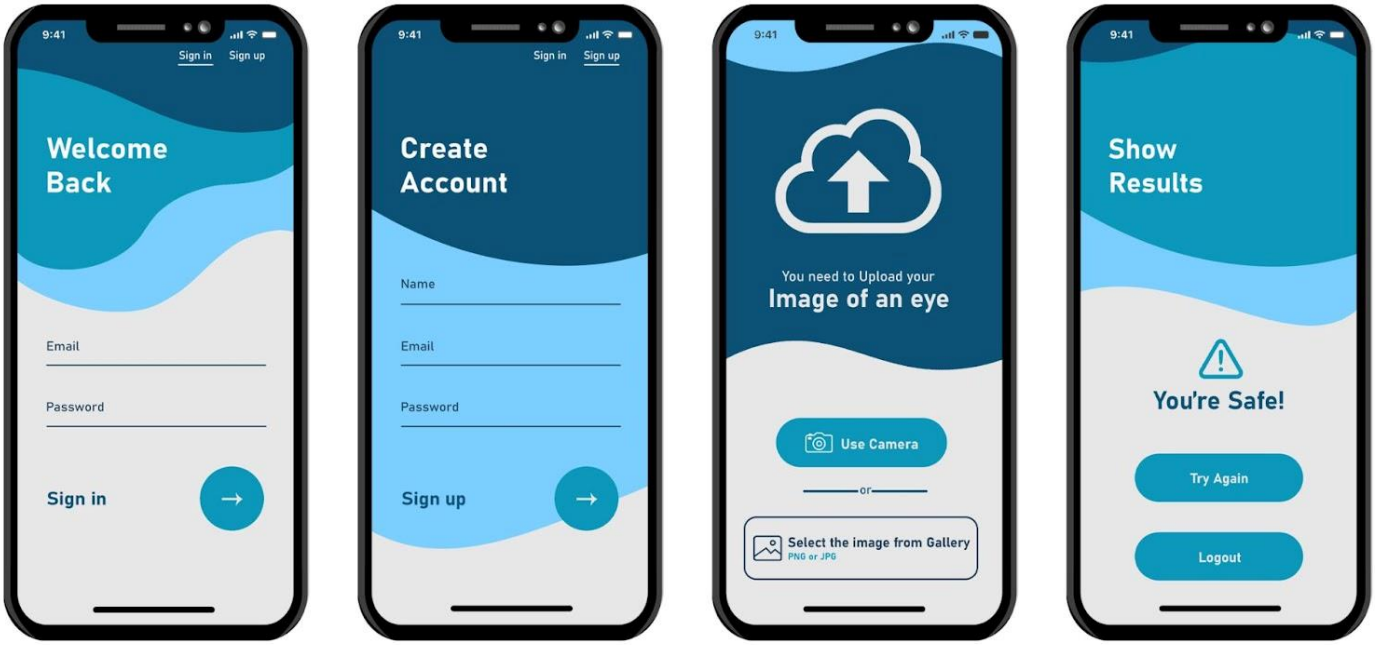


Figure 14 : UI Design - High Fidelity Prototype

6.4. Chapter Summary

This chapter clearly describes the layered architectures, and the System design were given more understanding using Class diagram with Attributes, a sequence diagram which explains the system sequentially and a flow chart for the process order of the project and some UI designs for the main Use Cases considered in the research.

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Appendix A – Gantt Chart

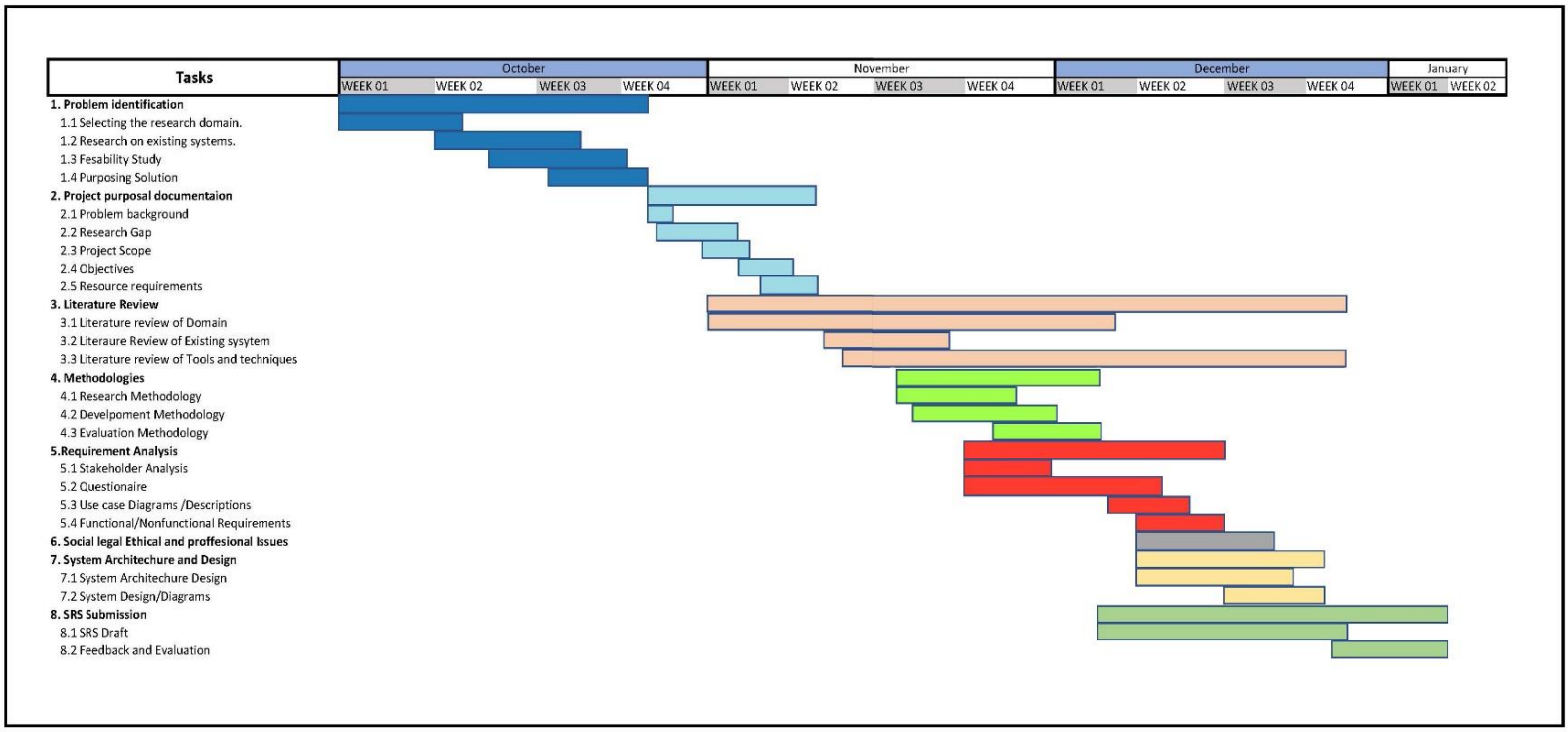
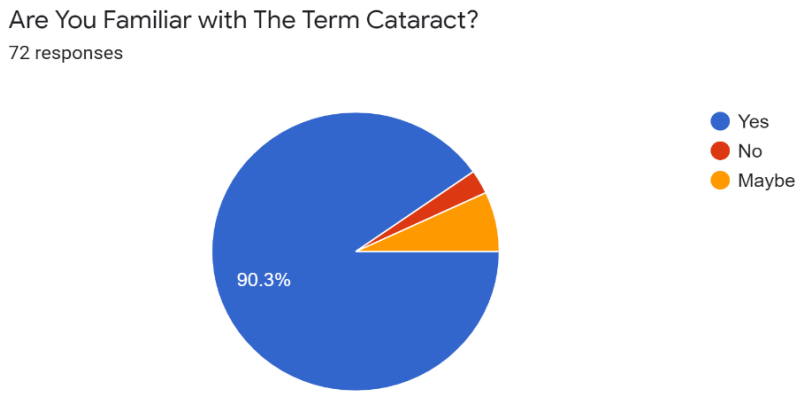


Figure 15 : Gantt Chart

Appendix B - Questionnaire Findings and Analysis

Question	Are You Familiar with The Term Cataract?
Question Aim	To identify awareness of the term known as cataract

Results



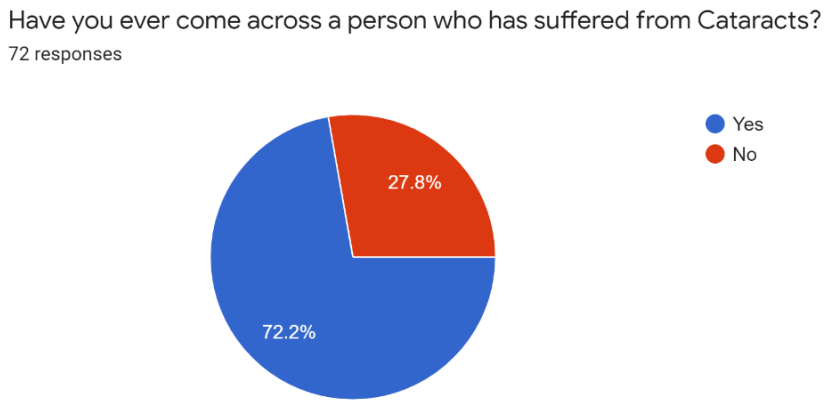
- A total of 90.3% of the participants are familiar with term Cataract and 6.9% are unaware about the term Cataract

Conclusion

The majority of people in Sri Lanka are familiar with the term Cataract, while a very small percentage are unaware of Cataracts. And yet another group of people who have a slight understanding of what Cataracts are.

Question	Have you ever come across a person who has suffered from Cataracts?
Question Aim	To identify how many people have experienced or seen cataracts

Results

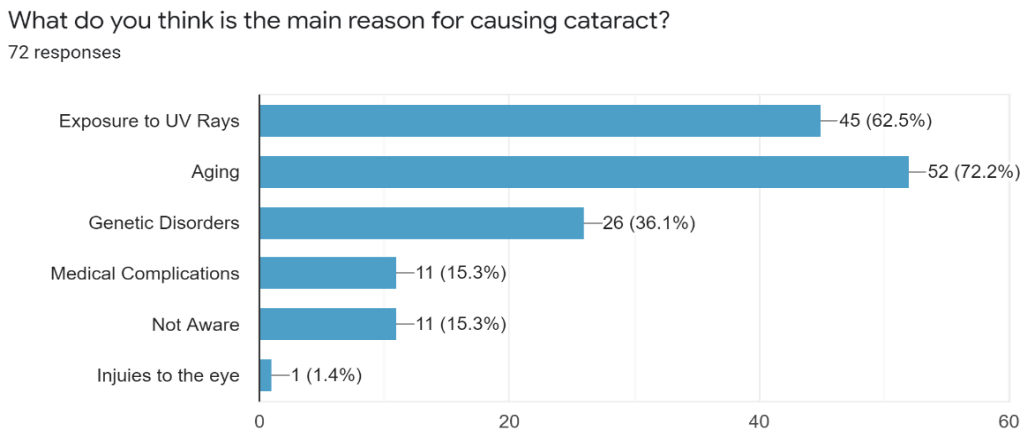


- A total of 72.2% have seen a person suffering from cataracts while 27.8% of the respondents have never seen.

Conclusion-
The majority of respondents to the survey either know or have seen a person suffering from cataracts, while the remaining percentage do not know anyone suffering from Cataracts.

Question	What do you think is the main reason for causing cataract?
Question Aim	To identify what the participants understanding is to what cataract is caused by

Results



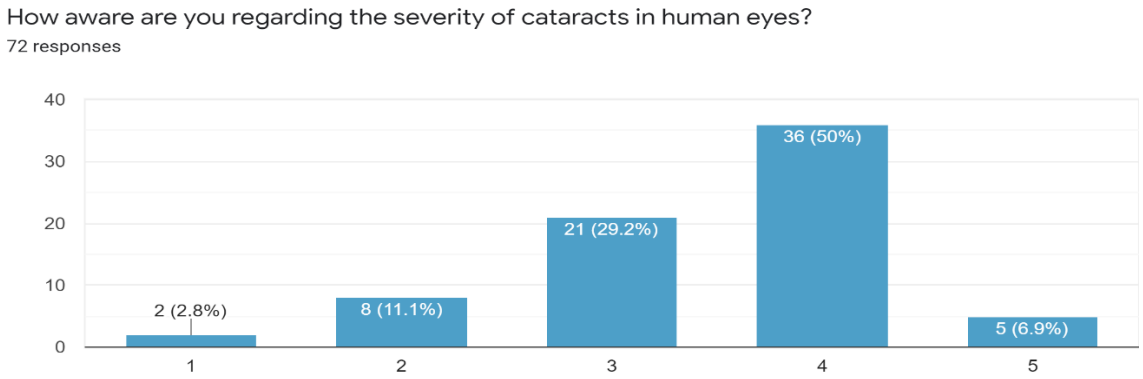
- 62.5% and 72.2 % of respondents believe exposure to UV Rays and aging are common causes, respectively, while 36. % believe it can be caused by genetic disorders, 15.3% are unaware of what causes cataracts and believe it is caused by medical complications, and finally, approximately 1.4% of respondents believe it is caused by eye injuries.

Conclusion-

Many people in Sri Lanka believe that the most common cause of Cataracts is aging and exposure to UV rays, while another group believes that it is also caused by medical complications, and there are very few who are unaware of what causes Cataracts. In conclusion, people in Sri Lanka believe that Cataracts is primarily caused by UV ray exposure, aging, genetic disorders, and medical complications.

Question	How aware are you regarding the severity of cataracts in human eyes?
Question Aim	To identify awareness of the severity of Cataract

Results



Results

- Around 50% of the respondents believe it is 4 on a scale of 1-5,29.2% of the responders think it is 3,11.1% of the responder's think it's a 2 and 6.9% think it is very severe being 5 and finally less than 3% of the responder's believe it is not severe at all.

Conclusion-

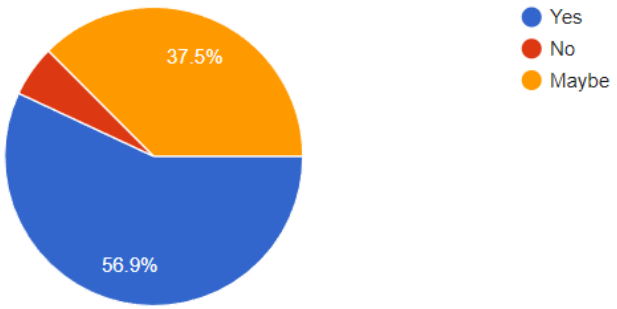
On a scale of 5, the majority of respondents believe Cataracts is severe, choosing 4 and 3, while 5 believe it is very severe, and approximately 8 respondents chose 2 and 2 respondents chose 1 as they believe it is not that severe. In conclusion, the majority of respondents believe that cataracts is a serious issue.

Question	Do you think that people in Rural areas of Sri Lanka are suffering from Cataracts more than the People in Urban areas?
Question Aim	Identify the awareness and compare the people who suffer from Cataracts in urban areas and rural areas

Results

Do you think that people in Rural areas of Sri Lanka are suffering from Cataracts more than the People in Urban areas?

72 responses



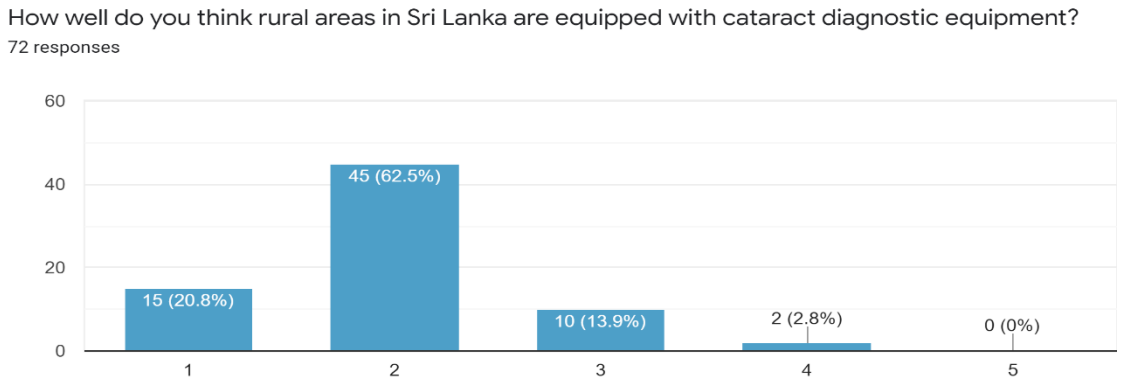
- 56.9% of the respondents have an opinion that in rural areas suffer from cataracts more than people in urban areas, while 37.5% of the respondents are unaware. And 5.6% of the respondent have the opinion that people in urban areas suffer from cataracts more than people in rural areas.

Conclusion-

This question is used to determine whether people in rural areas suffer from cataracts more than people in urban areas. In conclusion, the majority of respondents believe that people in rural areas suffer from cataracts more than people in urban areas, and only a minority believe that people in urban areas suffer from cataracts more than people in rural areas, while the remainder are unsure or have no idea.

Question	How well do you think rural areas in Sri Lanka are equipped with cataract diagnostic equipment?
Question Aim	To identify awareness of how well rural areas are equipped, for diagnosis of Cataracts

Results



- 2.8% of the respondents thinks they are very well equipped while majority of the respondents 62.5% believe they are slightly equipped. And 20.8% think they are very poorly equipped.

Conclusion-

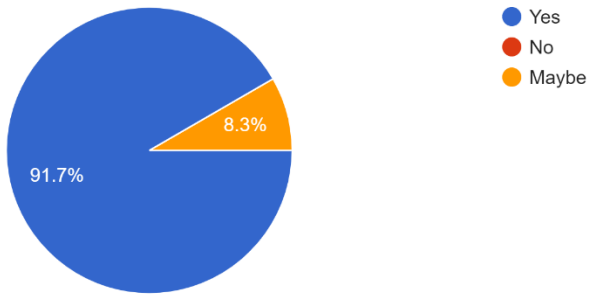
Majority of the people in Sri Lanka think that rural areas poorly quipped with Cataract diagnostic equipment and very litle believe that they are very well equipped proving that people are significantly aware of how rural areas are equipped with Cataract diagnostic equipment.

Question	Do you think that people in rural areas in Sri Lanka can benefit from a system which uses smartphone images to predict cataract stages? Since they have limited access to medical facilities.
Question Aim	To identify proposed projects relevance

Results

Do you think that people in rural areas in Sri Lanka can benefit from a system which uses smartphone images to predict cataract stages? Since they have limited access to medical facilities.

72 responses



- 91.7% of respondents have an opinion that the proposed system would be beneficial and 8.3% of the respondents are not sure.

Conclusion-

According to the results, the majority of respondents believe that the proposed system will benefit people in rural areas of Sri Lanka, while only a few are unsure, and none believe that the proposed system will be useless.

Appendix C - Usage of Project Management and Collaboration Software

1. Trello

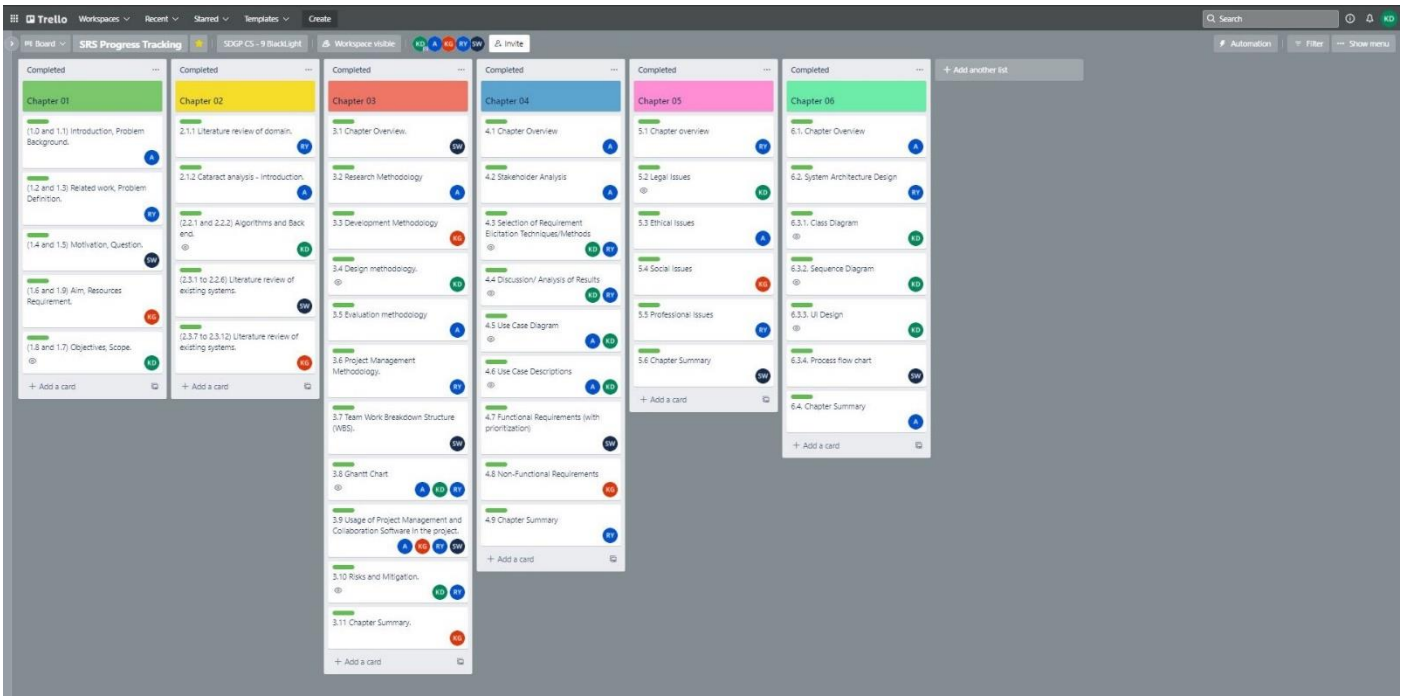


Figure 16 : Trello Board

2. Google Meet logs

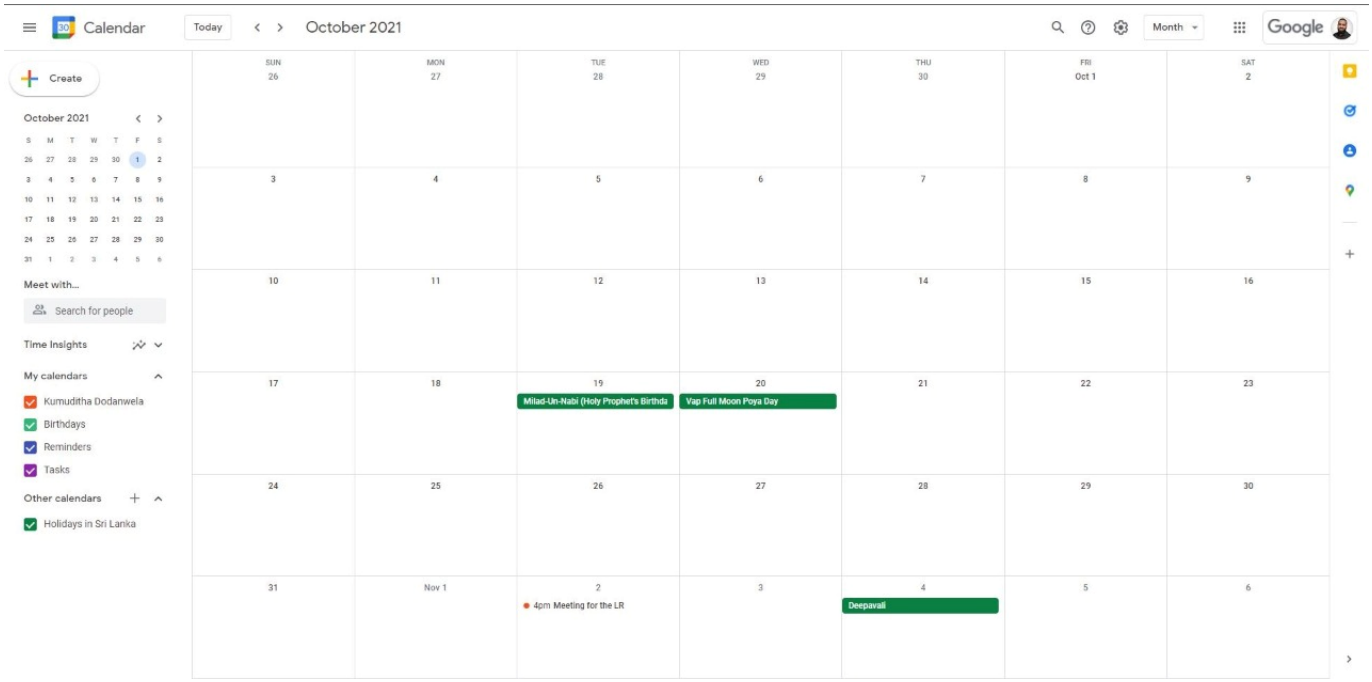


Figure 17 : Google Meet Log- October

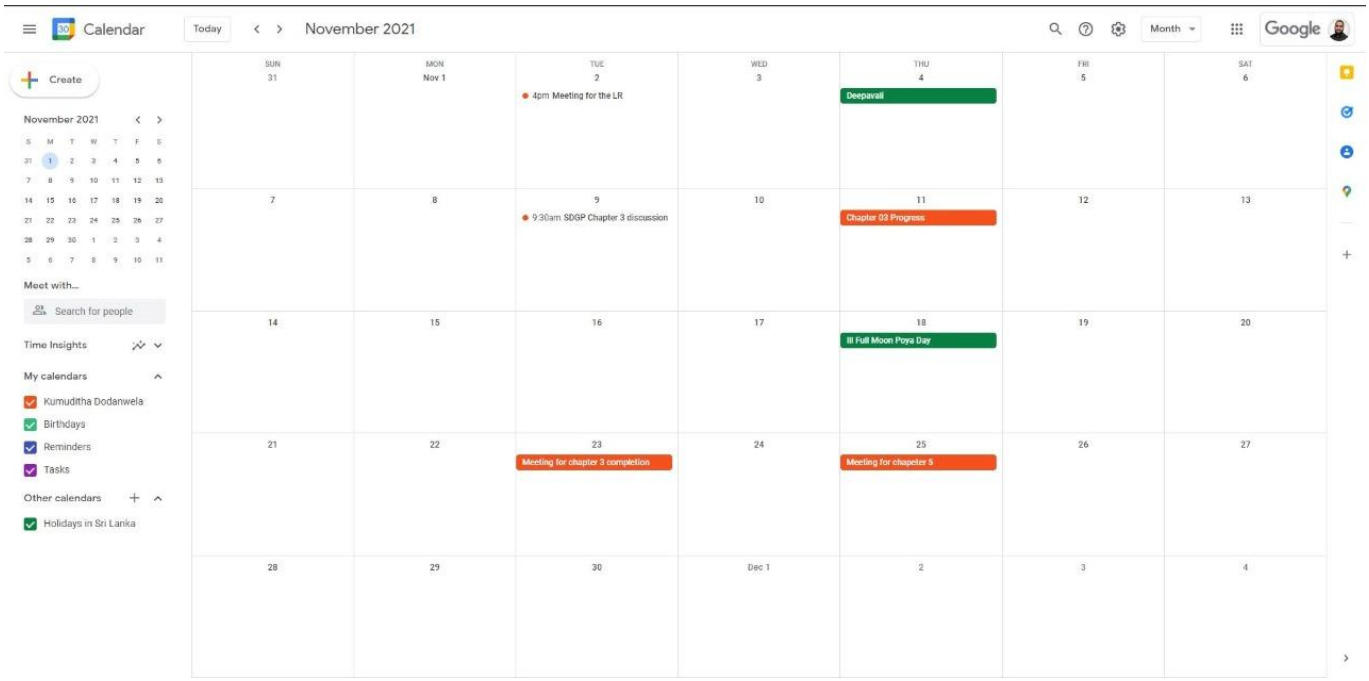


Figure 18 : Google Meet Log- November

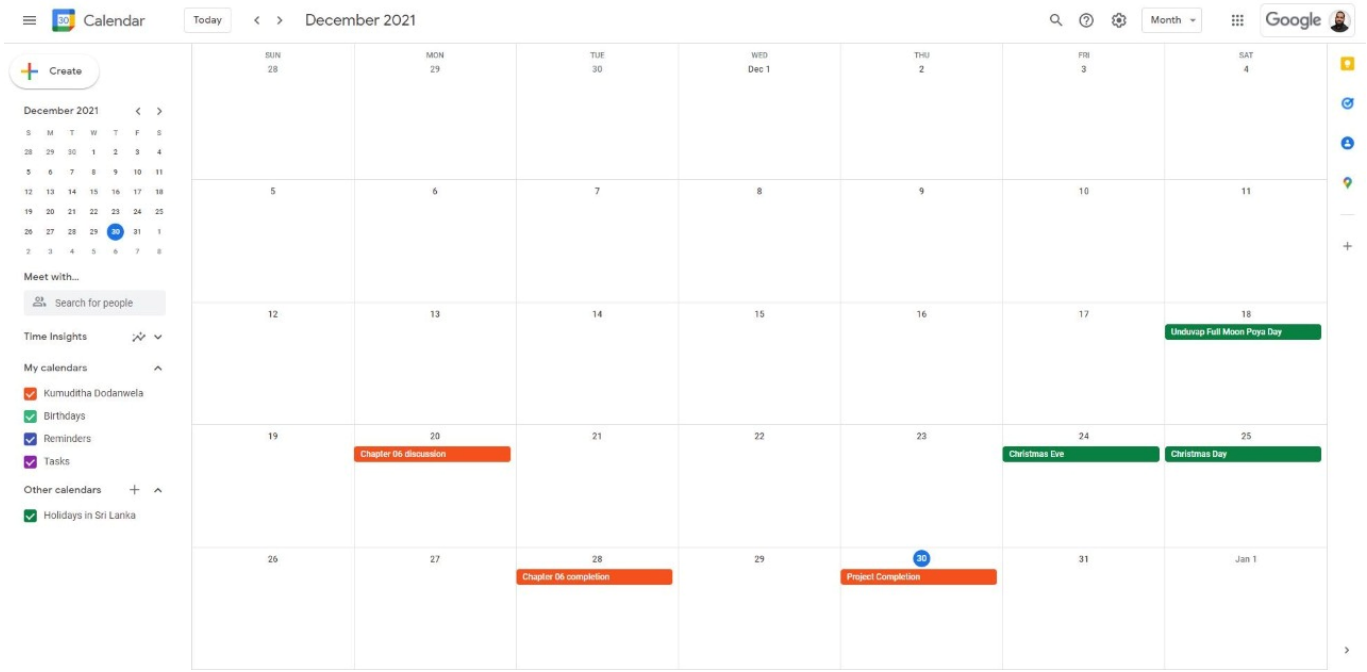


Figure 19 : Google Meet Log- December

Slack

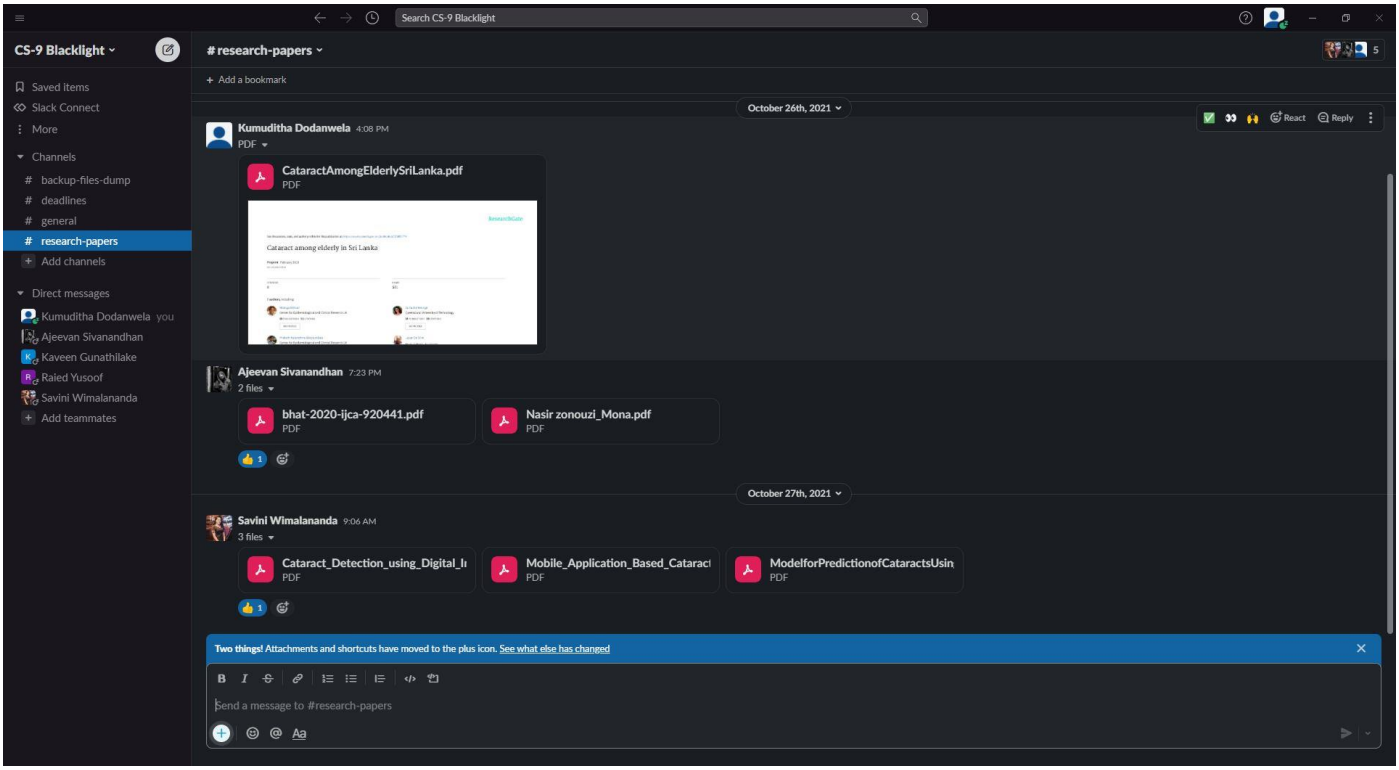


Figure 20 : Slack for Collaboration